Bachelor of Science in Mechatronics Engineering

Senior Design Graduation Project Report

Intelligent EEG Analysis for Binary Classification of Epileptic Seizures

June 1st, 2026

G1

Report by

Tala Saadeh	0216155
Raya Abul Feilat	0217161
Sara Makahleh	2210957

Supervisor

Dr. Mohammad Al Mashagbeh

DECLARATION STATEMENT

We, the undersigned students, confirm that the work submitted in this project report is entirely our own and has not been copied from any other source. Any material that has been used from other sources has been properly cited and acknowledged in the report.

We are fully aware that any copying or improper citation of references/sources used in this report will be considered plagiarism, which is a clear violation of the Code of Ethics of University of Jordan.

In addition, we have read and understood the legal consequences of committing any violation of the University's Code of Ethics.

	Student Name	Student ID	Signature	Date
1	Tala Saadeh	0216155	Tala Saadeh	01/03/2026
2	Raya Abul Feilat	0217161	Raya Abul Feilat	01/03/2026
3	Sara Makahleh	2210957	Sara Makahleh	01/03/2026

ABSTRACT

Epilepsy is a neurological disorder characterized by recurrent seizures caused by abnormal electrical activity in the brain. Electroencephalography (EEG) is a non-invasive technique used to record brain signals and is widely utilized in the diagnosis and monitoring of epilepsy. Detecting seizures from EEG recordings, however, can be time-consuming and requires expert interpretation. Therefore, automated seizure detection systems have become an important research area in biomedical engineering and artificial intelligence.

This report presents an intelligent EEG-based seizure classification system developed using the Temple University Hospital (TUH) EEG database. The proposed system focuses on designing, training, and evaluating a deep learning model capable of analyzing EEG signals and automatically detecting if a seizure is present. The EEG data undergoes appropriate preprocessing steps to improve signal quality and prepare the recordings for model training. The developed model learns to extract relevant patterns and features directly from the EEG data, enabling accurate and efficient seizure detection.

The primary objective of this project is to build a reliable and practical AI-based classification framework within the scope and timeline of an undergraduate graduation project. Emphasis is placed on achieving strong classification performance while maintaining a clear and structured system design.

ACKNOWLEDGEMENTS

We would first like to express our deepest gratitude to Almighty God for granting us the strength, patience, and perseverance to complete this graduation project successfully.

Our sincere appreciation goes to our project supervisor, Dr. Mohammad Al Mashagbeh, for his continuous guidance, support, and valuable feedback throughout the development of this project. His direction and encouragement played a fundamental role in shaping the quality and progress of our work.

Special thanks are also extended to Dr. Adham Al Sharkawi for his insightful contributions and strong expertise in artificial intelligence, which significantly enhanced the technical depth of the project.

We also extend our appreciation to Umniah Co. for their support in Python programming and cloud-related guidance, which contributed to the practical aspects of this work.

We would like to sincerely thank Dr. Ala'eddin Abul-Feilat, Neurologist, for his support and expert guidance on the medical aspects of this research. His valuable insights into seizure disorders greatly contributed to our understanding of the clinical domain and the development of this work.

In addition, we would like to thank the Department of Mechatronics Engineering and the University of Jordan for providing the academic environment, resources, and facilities necessary to complete this project.

Finally, we are deeply grateful to our families for their continuous encouragement, patience, and unwavering support throughout our academic journey.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

GLOSSARY

ABBREVIATION	DESCRIPTION
AI	Artificial Intelligence
AUC	Area Under the Curve (Performance Metric)
BGRU	Bidirectional Gated Recurrent Unit
BLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
DWT	Discrete Wavelet Transform
EEG	Electroencephalography
FFT	Fast Fourier Transform
GRU	Gated Recurrent Unit
IoT	Internet of Things
KNN	K-Nearest Neighbors
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MNE	Multimodal Next-Gen Enterprise (Python library for EEG processing)
PMLR	Proceedings of Machine Learning Research
RNN	Recurrent Neural Network
SGCN	Sequential Graph Convolutional Network
SVM	Support Vector Machine
TUH	Temple University Hospital EEG Database
GUI	Graphical User Interface
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
CSS	Cascading Style sheet
JS	JavaScript
API	Application Programming Interface
JSON	JavaScript Object Notation
HTTP	HyperText Transfer Protocol
AWS	Amazon Web Services
EC2	Elastic Compute Cloud

Table 1: Project Glossary

Chapter 1 INTRODUCTION

1.1 Background

Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures that affect approximately 50 million people worldwide according to the World Health Organization. Seizures themselves are transient episodes of abnormal, synchronous, and excessive electrical activity within the brain's neurons. These abnormal electrical discharges disrupt normal brain function and can manifest outwardly as a wide variety of clinical symptoms. Depending on the specific type of seizure and the areas of the brain involved, an episode can range from sudden, severe physical convulsions and sustained muscle stiffening to subtle, brief lapses in consciousness or awareness. Electroencephalography (EEG) is the primary non-invasive clinical tool used to monitor brain activity and detect seizure-related abnormalities. However, manual interpretation of EEG recordings is time-consuming and requires significant clinical expertise, particularly with long-duration recordings and large datasets.

Recent advances in artificial intelligence (AI) have significantly improved automated EEG-based seizure detection. Deep learning approaches such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid architectures have demonstrated strong performance in capturing spatial and temporal characteristics of EEG signals [1], [5], [7], [8]. More advanced techniques, including attention mechanisms [1], graph-based and non-Euclidean models [7], tensor decomposition methods [2], and self-supervised learning frameworks [3], have further enhanced classification accuracy and representation learning. In addition, machine learning models such as Random Forest, SVM, and boosting-based approaches have shown high reported performance in seizure detection and related EEG analysis tasks [9], [10].

Large-scale clinical datasets, particularly those derived from real hospital environments, play a crucial role in validating these models. The Temple University Hospital EEG database has become one of the most widely used repositories for bench-marking seizure detection systems [5], [7], [10].

1.2 Problem Definition

Epileptic seizures are typically identified through manual analysis of EEG recordings by neurologists. This process is time-consuming and difficult to scale when dealing with large clinical datasets. In addition, EEG signals are complex, noisy, and seizure events occur much less frequently than normal brain activity, creating a class imbalance problem that can affect model performance.

To address these challenges, this project proposes an AI-based EEG classification system using the Temple University Hospital (TUH) dataset. The system will preprocess EEG signals while applying different imbalance handling techniques to improve the accuracy and reliability of binary-class seizure classification using 1D CNN.

1.3 Literature Review

Recent advances in AI have significantly improved EEG-based epileptic seizure detection. The paper "Attention-based Deep Convolutional Neural Network for Classification of Generalized and Focal Epileptic Seizures" [1] combines a multi-head self-attention mechanism with a deep convolutional neural network (CNN) to classify seven subtypes of generalized and focal epileptic seizures. It provides a classification framework based on EEG characteristics for seven epileptic seizure types across the generalized and focal epilepsies: focal onset non-motor, generalized onset non-motor, simple partial, complex partial, absent, tonic, and tonic-clonic seizures with a weighted accuracy of 0.921 and a weighted F1 score of 0.902, outperforming a CNN without attention (0.767).

Complementing this, tensor decomposition techniques have been applied to large-scale EEG datasets to extract meaningful and interpretable patterns of brain physiology. The study "Tensor Decomposition of Large-Scale Clinical EEGs" [2] represents EEG signals as multi-dimensional tensors, capturing spatial, temporal, and spectral correlations while reducing dimensionality. By uncovering hidden structures in EEG data, tensor decomposition provides physiologically meaningful features that can improve downstream tasks such as seizure detection, offering a bridge between data-driven modeling and interpretability.

In another line of research, "Domain-Guided Self-Supervised Learning for EEG Classification" [3] proposes a framework in which EEG signals are pre-trained using domain-specific self-supervised tasks before being fine-tuned for seizure classification. This approach improves model generalizability and performance, particularly when labeled data are limited, demonstrating that leveraging intrinsic EEG structure via self-supervision can boost deep learning outcomes in clinical EEG analysis.

Topological approaches have also been introduced for EEG analysis. In "Topological Feature Extraction for Interictal Epileptiform Discharges in EEG" [4], high-dimensional EEG signals are transformed into low-dimensional phase space trajectories, and topological features are extracted to classify interictal epileptiform discharges. This method captures nonlinear and dynamic brain activity patterns often missed by traditional spectral or time-domain analyses, providing a complementary and interpretable feature set for seizure detection.

In another work, "Brain Age Prediction Using TUAB EEG Dataset" [5] uses raw EEG recordings from the Temple University Abnormal EEG (TUAB) dataset, focusing mainly on seizure patients who often have a neurodegenerative history. A deep learning model based on recurrent neural networks, specifically LSTMs, is developed to classify and predict patients' brain age. The model achieves 90% accuracy across six age groups and a mean absolute error of 7 years in brain age regression.

The paper "Graph-Based Non-Euclidean Approaches for EEG Analysis" [6] reviews graph-based, non-Euclidean techniques for EEG analysis, such as visibility graphs and graph neural networks, highlighting their advantage in capturing the nonlinear, dynamic, and structural properties of brain signals. It concludes that these approaches provide more biologically realistic and accurate representations of EEG data than traditional Euclidean methods.

AI-enabled wearable systems are reviewed as portable, non-invasive alternatives to bulky clinical tools. The survey "AI-Enabled Wearable Sensors for Real-Time Physiological Monitoring" [7] focuses on combining wearable sensors with AI models (CNN, RNN, LSTM) to analyze multi-modal physiological signals (ECG, EEG, EMG, motion, sweat, and temperature) in real time. The included studies, ranging from small clinical cohorts to large EEG/ECG databases, apply AI for filtering, feature extraction, and classification tasks such as arrhythmia detection, gait analysis, and sign-language translation. Reported accuracies often exceed 90% with strong sensitivity, specificity, robustness, and high user comfort. These systems enable early disease detection (e.g., Parkinson's, epilepsy), personalized therapy, and improved patient compliance, providing both theoretical and practical direction for future intelligent health monitoring.

A hybrid deep learning framework "Sequential Graph Convolutional Network for Automated Epileptic Seizure Detection" [8] for automated epileptic seizure detection from EEG signals was proposed by Jibon et al. (2024). It combines a Deep Recurrent Neural Network (DeepRNN) with a Sequential Graph Convolutional Network (SGCN). The method obtained 99.007% and 98.08% accuracy on the CHB-MIT and TUH datasets, respectively, by extracting spatial and temporal features using graph-based and recurrent modeling. The model demonstrated better generalization and robustness than traditional CNN, LSTM, and SVM-based methods.

In more recent years, "Multidimensional Transformer-LSTM-GRU Hybrid for Seizure Prediction" [9] proposed a hybrid model combining a multidimensional transformer with LSTM and GRU networks for the prediction of epileptic seizures. Tested on the CHB-MIT and Bonn EEG datasets, the model transforms EEG signals into spectrograms and uses multihead attention and recurrent layers to capture spatial-temporal features. It achieved high sensitivity, specificity, and F1 scores, outperforming previous methods.

Complementing this, "IoT and Cloud Computing-Based Automatic Epileptic Seizure Detection Using Random Forest Classifiers" [10] uses SVM, KNN, and Random Forest classifiers on preprocessed EEG data. Among the classifiers, Random Forest achieved the highest performance, with 99.89% accuracy, 100% recall, 99.73% precision, 99.80% specificity, and an F1 score of 99.86%. The system is designed to work within an IoT framework, enabling real-time monitoring and data transmission from wearable EEG devices to cloud-based analytics.

Moreover, in a machine learning framework, "Peri-Ictal EEG-Based Prediction of Postoperative Seizure Control" [11] created a machine learning framework that uses peri-ictal scalp EEG recordings to predict postoperative seizure control after temporal lobe resection. 294 Cleveland Clinic patients with drug-resistant temporal lobe epilepsy made up the dataset. The model significantly outperformed traditional clinical nomograms (C-index ≈ 0.65) by combining five minutes of peri-ictal EEG analysis (two minutes pre-ictal and three minutes post-ictal) with important clinical variables. The model achieved an AUC of 0.98 and testing accuracy above 90%.

Finally, using EEG recordings from the TUAB, "Brain Age Prediction Using Recurrent Neural Networks on TUAB EEGs" [12] examined the application of deep learning RNNs to predict brain age in seizure patients. Preprocessed EEG signals filtered using FFT and DWT methods were used to train four RNN architectures: LSTM, GRU, BLSTM, and BGRU. The top-performing model had a mean absolute error (MAE) of 7 years for brain age regression and 90% classification accuracy across six age groups.

Akor et al. (2025) proposed a hierarchical deep learning framework for epileptic seizure analysis using the TUSZ dataset. Their cascaded architecture combines a binary seizure detector with a multi-class seizure type classifier, achieving 99.40% detection accuracy and 99.90% seizure type classification accuracy while effectively addressing severe class imbalance through high-confidence filtering and class-weighted optimization.

1.4 Aims and Objectives

Our Aim is to develop a reliable and efficient AI-based classification system for epileptic seizures from EEG signals, validated on the Temple University Hospital (TUH) EEG dataset, to demonstrate accurate seizure classification within the scope of a graduation project.

Our Objective includes:

1. Collect and preprocess EEG data from the TUH database, including filtering, montage unification and segmentation of EEG recordings into fixed sized windows.

2. Calculate optimal weights for imbalance handling to optimize model performance.

3. Design and implement a deep learning model (1D CNN) capable of analyzing EEG signals for seizure classification.

4. Evaluate and validate the model using standard metrics (accuracy, sensitivity, specificity, F1-score) to ensure robust performance across unseen patients.

1.5 Project Plan

In this section, a Gantt Chart is presented outlining all of the tasks involved in the project, and their order, shown against a timescale.

#	Task	Responsibility	Weeks													
			1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Literature review	Raya, Sara														
2	Standards and limitations	Raya, Sara														
3	Problem Understanding and background study	Tala														
4	Dataset investigation	Tala, Raya, Sara														
5	Prepare and preprocess the EEG dataset for model training	Tala, Raya, Sara														
6	Design and implement the deep learning model for seizure detection.	Tala, Raya, Sara														
7	Train the model and evaluate its performance.	Tala, Raya, Sara														
8	Integrate the trained model with the cloud-based monitoring system and finalize the project.	Tala, Raya, Sara														

Table 2: Project Plan Log Book

1.6 Report Organization

This report is organized into five chapters, structured to guide the reader through the problem background, methodology, validation, and future directions of the EEG-based seizure detection project.

Chapter 1 – Introduction

1.1 Includes background information on epilepsy and EEG-based seizure detection.

1.2 Defines the problem statement, and research goal.

1.3 Provides an overview of relevant literature addressed by this work.

1.4 Presents the aims and objectives of the project.

1.5 Outlines the project plan, report structure, and chapter organization.

Chapter 2 – Dataset Description

2.1 Overview of EEG signals and why automated seizure classification is significant.

2.2 Describes of the TUH Seizure Corpus and the files hierarchy.

2.3 Addresses the challenges of the TUSZ Dataset.

2.4 System pipeline and an overview of the sequential steps taken to train the model.

Chapter 3 – System Overview

3.1 Addresses the Downscaling and sub-setting of the full TUSZ Dataset.

3.2 Presents the signal preprocessing steps in preparation for training.

3.3 Describes the 1D CNN model proposed for training

3.4 Outlines the steps taken to train and evaluate the full TUSZ Dataset.

Chapter 4 – Results and Discussion

4.1 Showcases the evaluation framework and performance metrics.

4.2 Outlines the class specific performance (seizure vs. non-seizure).

4.3 Presents the confusion matrix and error analysis.

4.4 Includes the results' justification and performance gaps.

Chapter 5 – System Integration and Software Architecture

5.1 Provides a high-level view of the system and its main components.

5.2 Describes the overall software structure.

5.3 Explains how data, models, and resources are loaded and managed during execution.

5.4 Defines required input data format, validation rules, and the expected preprocessing.

5.5 Presents the complete pipeline from EEG data input to seizure prediction output.

5.6 Discusses major implementation challenges encountered and the solutions adopted.

5.7 Outlines how the system and be scaled for real-world deployment.

5.8 Project summary

5.9 Highlights the project's main contributions and final conclusions.

Chapter 6 – Conclusion and Future Work

6.1 Summarizes the main findings and contributions of the project.

6.2 Highlights lessons learned, limitations, and areas for further enhancement.

6.3 Suggests directions for future work, including improved real-time performance.

Chapter 2 DATASET DESCRIPTION

This chapter presents the dataset and overall system framework used in this project for binary-class epileptic seizure classification. The study utilizes the Temple University Hospital EEG Seizure Corpus (TUSZ) Version 2.0.3, one of the most widely used clinical EEG datasets for seizure analysis. Due to the complexity of EEG signals, the dataset introduces several challenges, including severe class imbalance, noise, and variability between patients and recording sessions. Therefore, a preprocessing and data preparation pipeline was developed to standardize the EEG recordings and prepare them for deep learning model training. This chapter also provides an overview of the proposed system pipeline, including data acquisition, preprocessing, segmentation, dataset preparation, and model development stages.

2.1 Overview of EEG-Based Seizure Classification

2.1.1 EEG Signals and Epileptic Seizures

Electroencephalography (EEG) is the technique used to record the electrical activity of the brain through electrodes placed on the scalp. EEG signals are widely used in clinical diagnosis and neurological monitoring due to their ability to capture brain activity in real time. In patients with epilepsy, abnormal electrical discharges in the brain can lead to epileptic seizures, which often appear as distinctive patterns within EEG recordings. These patterns may vary depending on the type, duration, and location of the seizure activity.

EEG signals are highly non-stationary and may vary significantly between patients and seizure events. Seizure activity can appear as abnormal rhythmic discharges, spikes, or sharp waveforms with varying temporal and frequency characteristics. This variability increases the complexity of automated seizure detection and requires models capable of learning robust temporal patterns directly from multichannel EEG recordings.

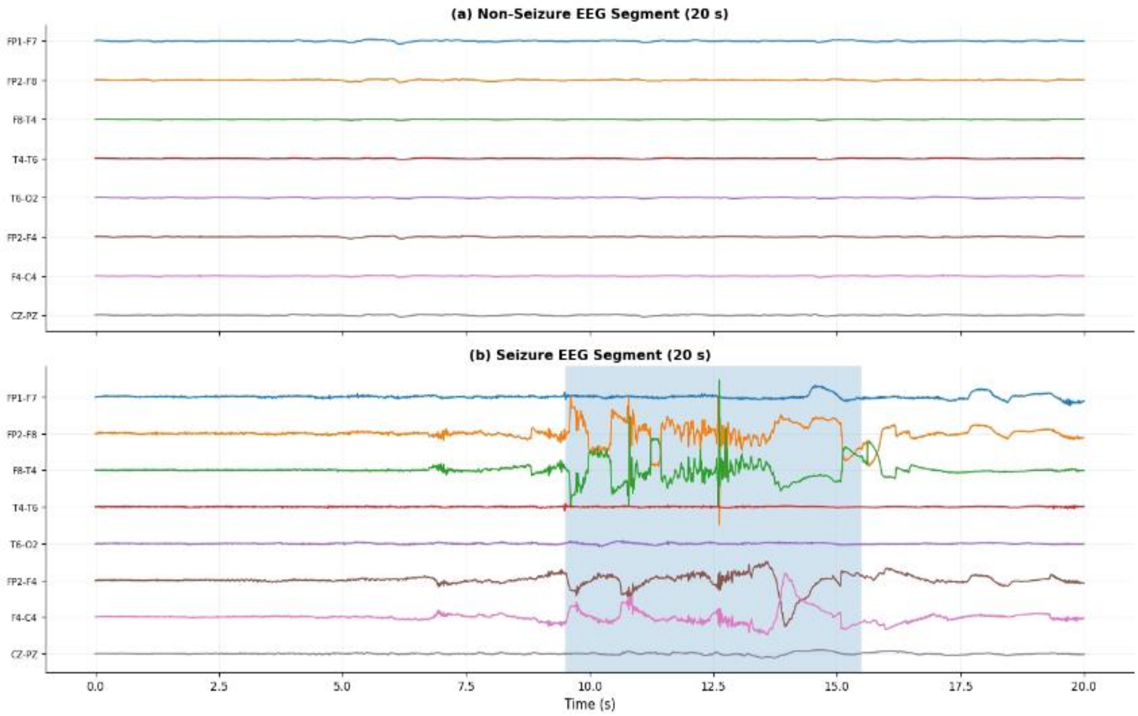

Figure 1: Seizure vs Non-seizure EEG channel signals

2.1.2 Automated Seizure Classification

Recent advances in artificial intelligence and deep learning have enabled the development of automated systems capable of detecting and classifying epileptic seizures from EEG recordings. Deep learning models can learn complex temporal and spatial patterns directly from EEG signals without relying heavily on manual feature extraction techniques.

Automated seizure classification systems can support clinicians by reducing the time required for EEG interpretation and assisting in continuous patient monitoring. In addition, these systems have the potential to improve the consistency of seizure detection and contribute to the development of intelligent healthcare and real-time monitoring applications.

In this project, convolutional neural networks (CNNs) were selected due to their ability to efficiently learn local temporal features from multichannel EEG recordings while maintaining lower computational complexity compared to more advanced transformer-based architectures. Conv1D layers are particularly suitable for time-series biomedical signals because they can capture seizure-related temporal patterns directly from sequential EEG windows.

2.1.3 Challenges of EEG-Based Classification

Despite recent progress in automated EEG analysis, seizure classification remains a challenging task due to the complex nature of EEG signals and the variability of clinical data. EEG recordings are often contaminated by artifacts and noise caused by muscle movement, eye blinking, electrode interference, and external electrical activity. Furthermore, seizure characteristics may differ significantly between patients and across seizure categories, making model generalization difficult.

Another major challenge is the severe class imbalance commonly found in clinical EEG datasets. Non-seizure or background activity typically represents most of the recordings, while some seizure classes contain only a limited number of samples. This imbalance can negatively affect model training and lead to biased predictions toward majority classes. Consequently, effective preprocessing, data preparation, and imbalance handling strategies are essential for developing reliable seizure classification systems.

Another important challenge is achieving patient-independent generalization. EEG seizure patterns may differ significantly across patients in waveform morphology, duration, amplitude, and frequency characteristics. Consequently, models trained on specific patients may fail to generalize effectively to unseen recordings, making patient-disjoint dataset splitting essential for reliable evaluation.

2.2 Temple University Seizure Corpus (TUSZ)

2.2.1 Dataset Overview

The data set used in this project is the Temple University Hospital EEG Seizure Corpus (TUSZ) Version 2.0.3. TUSZ is considered one of the largest used publicly available clinical EEG datasets for epileptic seizure analysis. The dataset contains EEG recordings collected from multiple patients in real clinical environments and includes expert annotations identifying the start and end times of seizure events. These annotations enable accurate identification of seizure and non-seizure segments during data preparation.

The TUSZ dataset was developed to support the development and evaluation of automated seizure detection and classification systems. Due to its large scale, diversity of seizure types, and clinically realistic EEG recordings, the dataset provides a valuable benchmark for developing state-of-the-art artificial intelligence models for epileptic seizure analysis.

2.2.2 EEG Recording Structure

The EEG recordings in the TUSZ dataset are stored using the European Data Format (EDF), which is a widely used standard format for storing multichannel physiological and biomedical signals. The dataset is organized into three main subsets: training, development, and evaluation. The training set is used for model training, while the development set is used for validation and hyperparameter tuning during the experimentation phase. The evaluation set is considered a blind test set and is reserved for final model assessment after the training and tuning processes are completed. This separation helps ensure unbiased evaluation of the model's generalization performance.

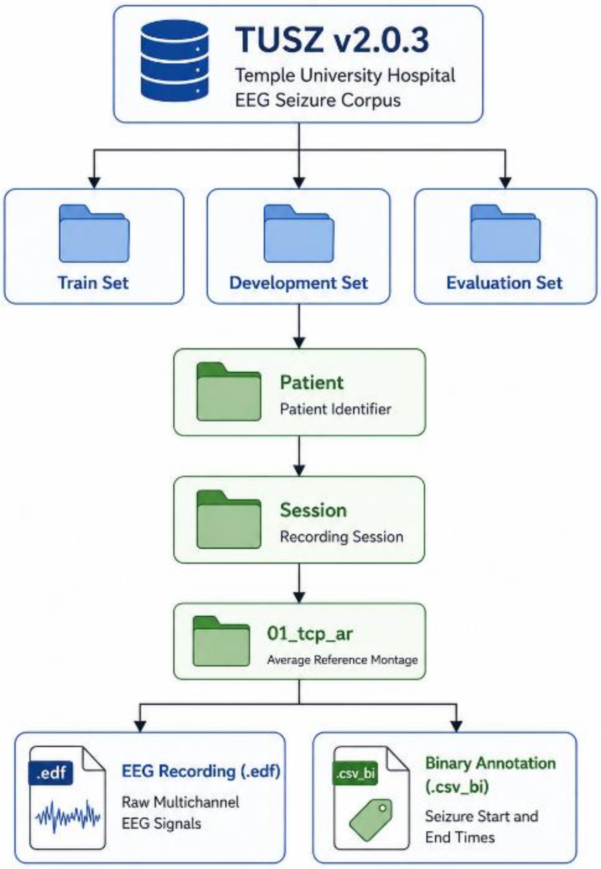

Figure 2: TUHZ v2.0.3 File Hierarchy

Within the dataset structure, each patient is represented by an anonymized directory containing one or more recording sessions. Each session may contain multiple EEG recordings with variable durations depending on the clinical examination. This hierarchical organization reflects the real clinical structure of EEG acquisition and allows the dataset to capture significant variability between patients and recording conditions.

The EEG recordings follow the international 10–20 electrode placement system, which is commonly used in clinical EEG monitoring. The dataset includes recordings acquired using both Linked Ears (LE) and Average Reference (AR) montages. In this project, only recordings using the AR montage were selected to maintain consistency during preprocessing and model training.

In addition to the EEG recordings, the dataset provides annotation files in both CSV and binary CSV formats for each recording. These annotation files contain detailed information about seizure events, including seizure type, start time, and end time. The provided annotations were used during the preprocessing phase to generate labeled EEG segments for supervised model training and evaluation.

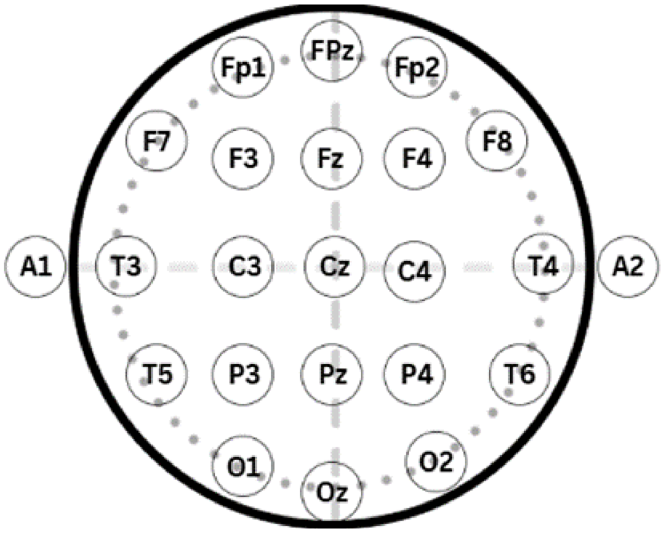

Figure 3: Electrode placement based on the 10-20 system and the corresponding TCP montage used in the TUSZ dataset

2.2.3 Seizure Categories

Although the TUSZ dataset contains multiple annotated seizure categories, this project focuses on binary seizure detection rather than seizure-type classification. Therefore, all seizure categories, including focal, generalized, tonic, tonic-clonic, and absence seizures, were merged into a single seizure class during preprocessing, while normal background EEG activity was assigned to the non-seizure class. This binary mapping simplified the classification task and allowed the model to focus on distinguishing abnormal epileptic activity from normal brain activity under highly imbalanced clinical conditions.

Class code	Event name	Description
BCKG	Background	Non-seizure EEG activity
FNSZ	Focal Non-Specific Seizure	Seizure originating from a localized brain region
GNSZ	Generalized Seizure	Seizure affecting both hemispheres simultaneously
CPSZ	Complex Partial Seizure	Focal seizure associated with impaired awareness
ABSZ	Absence Seizure	Brief seizure characterized by loss of awareness
TNSZ	Tonic Seizure	Seizure involving sustained muscle stiffening
TCSZ	Tonic-Clonic Seizure	Seizure involving tonic and clonic phases
MYSZ	Myoclonic Seizure	Seizure characterized by sudden muscle jerks

Table 3: Seizure Types in the TUH v2.0.3 data set

2.3 Dataset Challenges and Class Imbalance

2.3.1 Class Imbalance

A major challenge in developing robust deep learning systems for automated seizure detection is the severe class imbalance between seizure and non-seizure EEG activity. The Temple University Hospital EEG Seizure Corpus (TUSZ v2.0.3) spans approximately 1,473.4 hours of EEG recordings distributed across 7,364 files separated into patient-independent training, development, and evaluation subsets. However, the majority of the recorded EEG duration corresponds to normal background activity, while seizure events occupy only a small fraction of the recordings. This imbalance becomes even more significant after segmenting the EEG signals into fixed temporal windows, resulting in a substantially larger number of non-seizure samples compared to seizure samples.

Such imbalance can negatively affect model optimization by biasing the learning process toward the dominant background class. Consequently, a classifier may achieve high overall accuracy while still failing to correctly identify seizure events. Since seizure sensitivity is clinically more important than simply maximizing accuracy, imbalance-aware strategies were incorporated throughout the proposed framework. These included stratified subset generation, preservation of seizure-positive patients during down-scaling, and class-weighted training to encourage the model to learn seizure-related EEG patterns more effectively and improve generalization across unseen recordings.

	Total Patients	**Patients with Seizure**	**Total Sessions**	**Sessions with Seizures**	**Total Files**	**Files with Seizures**
Train	579	208	1,175	352 (29.9%)	4,667	873 (18.7%)
Dev	53	45	342	114 (33.3%)	1,832	325 (17.7%)
Eval	43	34	126	63 (50.0%)	865	195 (22.5%)

Table 4: Seizure vs No-Seizure Distribution in Patients, Sessions, and Files

2.3.2 Variability Between Patients and Recordings

EEG recordings exhibit significant variability across patients and recording sessions, which increases the complexity of automated seizure classification. Brain activity is highly dynamic, and EEG signals may contain various neurological patterns in addition to seizure-related activity. As a result, the characteristics of EEG recordings can differ substantially between patients depending on physiological conditions, seizure type, recording environment, and individual brain activity patterns.

Another important challenge is the variability within the seizure events themselves. The same seizure category may appear differently across patients in terms of waveform shape, signal amplitude, duration, and frequency characteristics. This variability makes it difficult for deep learning models to learn generalized seizure representations that perform consistently across unseen patients.

In addition to physiological variability, the EEG recordings in the dataset were not standardized in duration or signal quality. Recording lengths varied significantly between sessions and patients, resulting in inconsistent data sizes and seizure distributions across recordings. Furthermore, differences in channel quality and noise levels introduced additional challenges during model training and evaluation. These factors required careful preprocessing and patient-independent evaluation to ensure reliable classification performance across unseen EEG recordings.

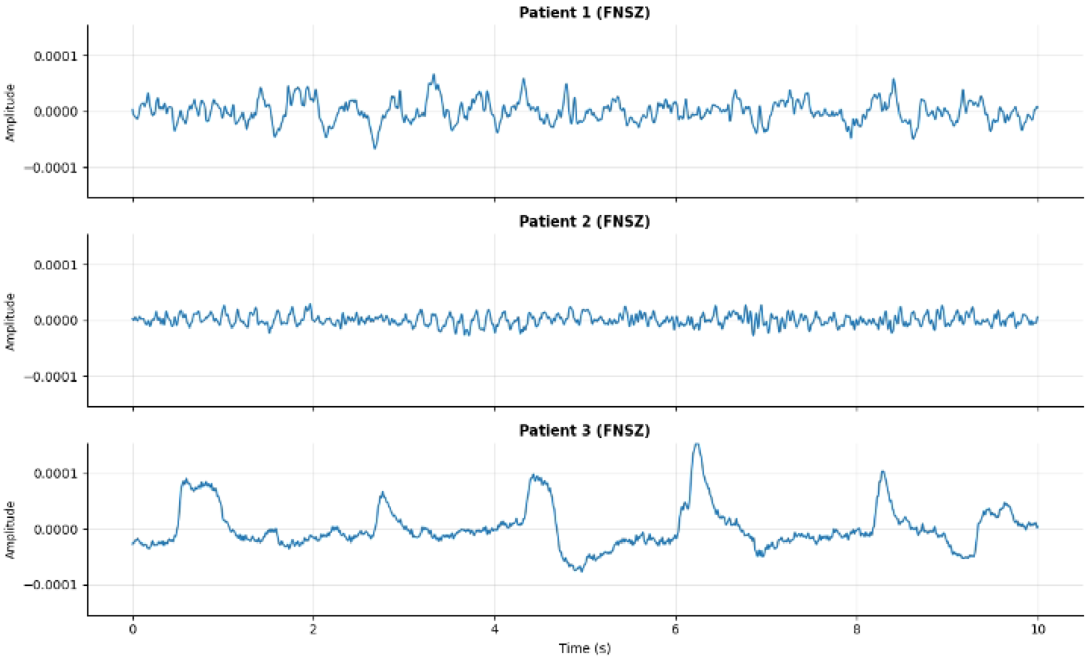

Figure 4: Seizure segment from different patients of the same seizure type

2.3.3 Noise and Artifacts in EEG Signals

EEG electrodes can also capture various physiological and non-physiological signals that alter the measured brain potential. These unwanted signal components are commonly referred to as artifacts. Artifacts may originate from eye blinking, muscle movement, patient motion, or poor electrode contact, and they can significantly distort the EEG waveform. In some cases, artifacts may produce sudden spikes or abnormal patterns that resemble epileptiform activity, increasing the difficulty of distinguishing real seizure events from noisy signal fluctuations.

In addition to physiological artifacts, EEG recordings are also affected by environmental and electrical noise sources. Common sources include power-line interference, electrode instability, sensor malfunction, and variations in recording quality between sessions. These disturbances can reduce signal clarity and introduce inconsistencies within the dataset, particularly in long-duration clinical recordings collected under different operating conditions.

The presence of noise and artifacts can negatively affect deep learning model performance by masking important seizure-related characteristics within the EEG signals. Consequently, models may learn irrelevant signal patterns instead of clinically meaningful seizure activity, leading to reduced generalization capability and increased false alarm rates. To reduce these effects, preprocessing techniques such as montage standardization, frequency filtering, and signal segmentation were applied before model training and evaluation.

2.4 System Pipeline

The proposed EEG seizure classification framework was implemented through a multi-stage processing pipeline designed to handle large-scale clinical EEG recordings efficiently while maintaining patient-independence evaluation.

• Phase 1: Dataset Reduction and Sub-setting
A reduced subset of the original TUSZ dataset was generated during the initial experimentation phase to simplify debugging, preprocessing validation, and hyperparameter tuning. The subset preserved the natural seizure-to-non-seizure distribution and maintained patient-independent separation between the training, development, and evaluation subsets.

• Phase 2: Signal Preprocessing and Window Generation
The EEG recordings were preprocessed using montage standardization and frequency filtering techniques to improve signal consistency and reduce noise artifacts. Only recordings acquired using the AR montage were selected. The EEG signals were then segmented into fixed temporal windows with corresponding binary seizure labels.

• Phase 3: Model Training and Validation
The segmented EEG windows were used to train a Conv1D-based deep learning model for binary seizure classification. Class-weighted training, validation monitoring, and early stopping strategies were incorporated to address the severe class imbalance problem and reduce overfitting during training.

• Phase 4: Full Dataset Deployment and GPU Training
After validating the preprocessing and training pipeline on the reduced subset, the framework was scaled to the full TUH EEG dataset. Preprocessed EEG windows were serialized into compressed NPZ chunks and streamed dynamically during training to reduce memory consumption and enable efficient handling of millions of EEG windows.

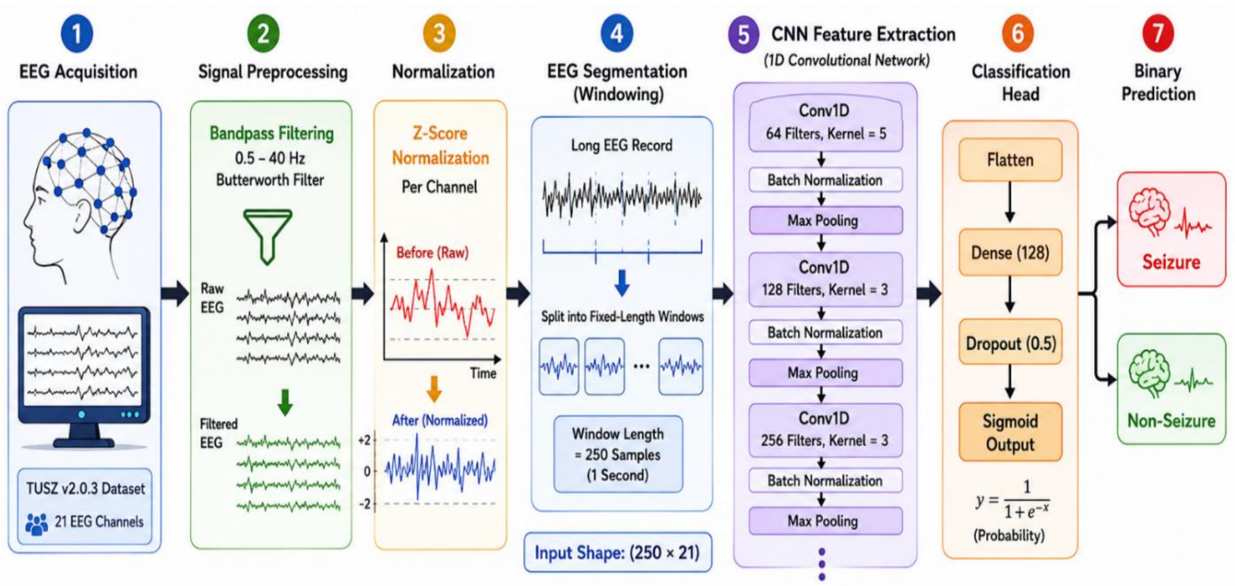

Figure 5: System Preprocessing and Training Pipeline

Chapter 3 SYSTEM OVERVIEW

3.1 Downscaling and Sub-Setting

3.1.1 Motivation for Dataset Reduction

The original Temple University Hospital Seizure Corpus (TUSZ v2.0.3) contains a large volume of multichannel clinical EEG recordings exceeding 81 GB in total storage size. Although the dataset provides significant diversity and clinical realism, directly preprocessing and training deep learning models on the full dataset during the initial experimentation phase introduced major computational challenges. These included excessive preprocessing times, large memory requirements, unstable runtime behavior, and prolonged debugging cycles that slowed the development process considerably.

During the early implementation stages, several preprocessing and training operations required repeated validation, including EDF parsing, annotation synchronization, montage conversion, binary label assignment, and window generation. Performing these operations directly on the complete dataset significantly increased execution time and complicated iterative testing. Therefore, a reduced subset of the dataset was initially generated to simplify debugging, accelerate experimentation, and validate the proposed preprocessing and training pipeline before scaling to the full dataset.

The reduced subset was designed to preserve the general statistical characteristics of the original dataset while maintaining patient-independent separation between the training, development, and evaluation subsets. Maintaining realistic seizure-to-non-seizure distributions during this phase was important to ensure that the preliminary experiments remained representative of the final classification problem and allowed meaningful evaluation of the proposed framework under limited computational resources.

3.1.2 Patient-Preserving Stratified Sub-setting

To ensure that the reduced dataset remained representative of the original TUSZ distribution, a patient-preserving stratified sub-setting strategy was implemented during dataset reduction. The subset generation process preserved the separation between seizure-positive and non-seizure patients while maintaining patient-independent boundaries across the training, development, and evaluation subsets. This prevented data leakage between subsets and ensured that recordings belonging to the same patient were not shared across different stages of the learning process.

Special attention was given to preserving seizure-positive recordings during the reduction process due to the severe imbalance between seizure and non-seizure EEG activity within the dataset. Since seizure events represent only a small fraction of the total recordings and segmented windows, random reduction could significantly decrease seizure representation and negatively affect model training. Therefore, the subset generation process maintained representative seizure distributions across all subsets while reducing the total number of recordings to a computationally manageable size.

After subset generation, statistical verification procedures were performed to confirm that the final reduced subsets preserved clinically meaningful seizure distributions and patient diversity. The resulting subsets were then used throughout preprocessing validation, model debugging, and preliminary training experiments before deployment on the complete dataset.

3.1.3 Initial Debugging and Experimental Validation

Before deploying the proposed framework on the complete dataset, multiple debugging and validation stages were conducted using the reduced subset to verify the correctness of the preprocessing and training pipeline. These experiments focused on validating EDF loading, annotation extraction, seizure interval synchronization, binary label mapping, and EEG window generation. Several intermediate outputs were manually inspected during preprocessing to ensure that seizure windows were correctly aligned with the corresponding EEG signal segments and that non-seizure background activity was properly represented.

The reduced subset also enabled rapid experimentation with model configurations, batch sizes, class weighting strategies, and training callbacks before scaling to the complete dataset. This iterative debugging phase significantly accelerated development and helped identify preprocessing inconsistencies, runtime limitations, and imbalance-related training behavior prior to full-scale deployment.

3.2 Signal Pre-Processing and Windowing

Following EEG recording selection, multiple preprocessing operations were applied to prepare the EEG signals for deep learning-based seizure classification. The EEG recordings were loaded from EDF files using the MNE-Python library with full signal preloading enabled to simplify downstream preprocessing operations. To ensure consistent multichannel input dimensions across all recordings, only the common 21 EEG channels shared between the selected recordings were retained during preprocessing.

After channel selection, all EEG recordings were resampled to a unified sampling frequency of 250Hz to standardize temporal resolution across the dataset and simplify fixed-window segmentation. A digital bandpass filter with cutoff frequencies of 0.5 Hz and 40 Hz was then applied using an IIR filtering approach to suppress low-frequency baseline drift and high-frequency noise while preserving clinically relevant EEG activity associated with seizure patterns.

Following filtering, z-score normalization was performed independently for each EEG channel by subtracting the channel mean and dividing by the corresponding standard deviation. This normalization stage reduced amplitude variability between recordings and improved signal consistency across patients and recording sessions. The preprocessing pipeline then extracted seizure annotations from the corresponding CSV annotation files and converted seizure start and stop times into sample-level intervals synchronized with the EEG recordings.

To generate standardized deep learning inputs, the EEG recordings were segmented into fixed temporal windows using a sliding-window approach. Each extracted EEG window was represented using a tensor shape of (250, 21), corresponding to 250 temporal samples across 21 EEG channels. Binary labels were assigned based on temporal overlap between each EEG window and the annotated seizure intervals. Windows overlapping seizure activity were assigned the seizure label (1), while all remaining windows were assigned the non-seizure label (0).

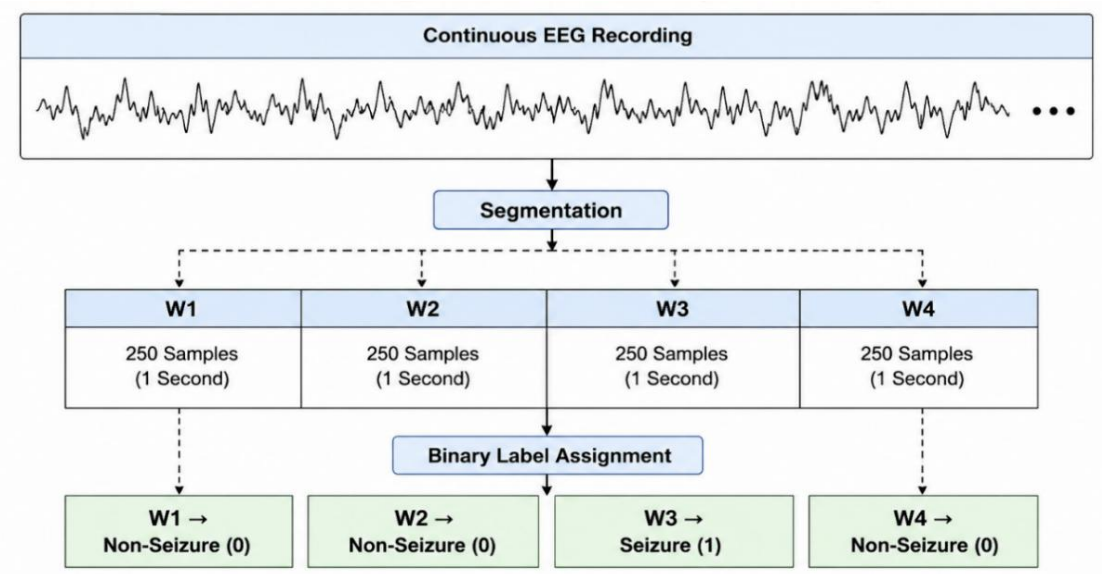

Figure 6: EEG Window Segmentation and Binary Label Assignment Process

The preprocessing pipeline was extensively validated during debugging and experimentation. Multiple intermediate EEG outputs and generated windows were manually inspected to verify correct annotation synchronization, signal integrity, and label assignment prior to large-scale dataset serialization and GPU-based model training.

Parameter	Value
Dataset	TUSZ v2.0.3
Selected Montage	01_tcp_ar
Number of EEG Channels	21
Input File Format	EDF
Target Sampling Frequency	250 Hz
Filter Type	Bandpass Filter
Lower Cutoff Frequency	0.5 Hz
Upper Cutoff Frequency	40 Hz
Normalization Method	Z-Score Normalization
Window Length	250 Samples
Window Duration	1 Second
Window Stride	250 Samples
Window Overlap	0%
Input Tensor Shape	(250, 21)
Classification Type	Binary Classification

Table 5: EEG Signal Preprocessing and Window Generation Parameters

3.3 Model Selection

3.3.1 Training Model

Selecting an appropriate deep learning architecture was an important step in developing the proposed EEG seizure classification framework. Since EEG recordings represent multichannel temporal biomedical signals, the selected model needed to efficiently capture local temporal seizure-related patterns while maintaining computational feasibility for large-scale dataset training. Convolutional Neural Networks (CNNs), particularly one-dimensional convolutional architectures (Conv1D), are widely used for time-series biomedical signal analysis due to their ability to automatically learn hierarchical temporal features directly from raw sequential data.

In this project, a Conv1D-based architecture was selected because of its ability to process EEG windows efficiently while maintaining lower computational complexity compared to more advanced architectures such as transformers and graph-based neural networks. Conv1D layers are particularly suitable for EEG seizure detection because they can capture localized temporal waveform characteristics associated with epileptic activity, including rhythmic discharges, spikes, and abrupt signal variations.

Another advantage of Conv1D architecture is their compatibility with large-scale clinical EEG datasets and GPU-based training environments. Compared to recurrent architectures such as LSTMs and GRUs, Conv1D models generally require lower training time and memory consumption while still achieving strong performance in time-series classification tasks. This made the architecture suitable for iterative experimentation, imbalance-aware training, and large-scale EEG window classification within the computational limitations of the project environment.

3.3.2 Proposed Conv1D Architecture

The proposed deep learning model was implemented using a one-dimensional convolutional neural network (Conv1D CNN) architecture designed specifically for binary EEG seizure classification. The architecture was developed to process fixed-size multichannel EEG windows and automatically learn seizure-related temporal patterns directly from the preprocessed EEG signals without relying on handcrafted feature extraction methods.

The input to the network consisted of EEG windows with a tensor shape of (250, 21), representing 250 temporal samples across 21 EEG channels. The feature extraction stage of the architecture consisted of three sequential convolutional blocks. The first Conv1D layer utilized 64 convolutional filters with a kernel size of 5 and ReLU activation to capture low-level temporal EEG patterns from the input signals. This was followed by batch normalization and max-pooling operations to stabilize feature distributions and reduce temporal dimensionality.

The second convolutional block increased the number of filters to 128 using a kernel size of 3, allowing the network to learn more complex seizure-related temporal representations. Similarly, the third convolutional block utilized 256 filters with a kernel size of 3 to extract higher-level EEG features associated with epileptic activity. Batch normalization and max-pooling layers were applied after each convolutional operation to improve convergence behavior and reduce computational complexity during training.

Following the convolutional feature extraction stage, the generated feature maps were flattened into a one-dimensional representation and passed into a fully connected dense layer containing 128 neurons with ReLU activation. A dropout layer with a dropout rate of 0.5 was incorporated before the final classification layer to reduce overfitting and improve model generalization across unseen patients. The output layer consisted of a single neuron with sigmoid activation to generate binary seizure probability predictions ranging between 0 and 1.

The complete architecture contained approximately 1.08 million trainable parameters and was optimized using GPU-based training within the TensorFlow and Keras deep learning framework. The proposed Conv1D architecture achieved an effective balance between classification performance and computational efficiency, making it suitable for large-scale EEG window classification and iterative experimentation throughout the project.

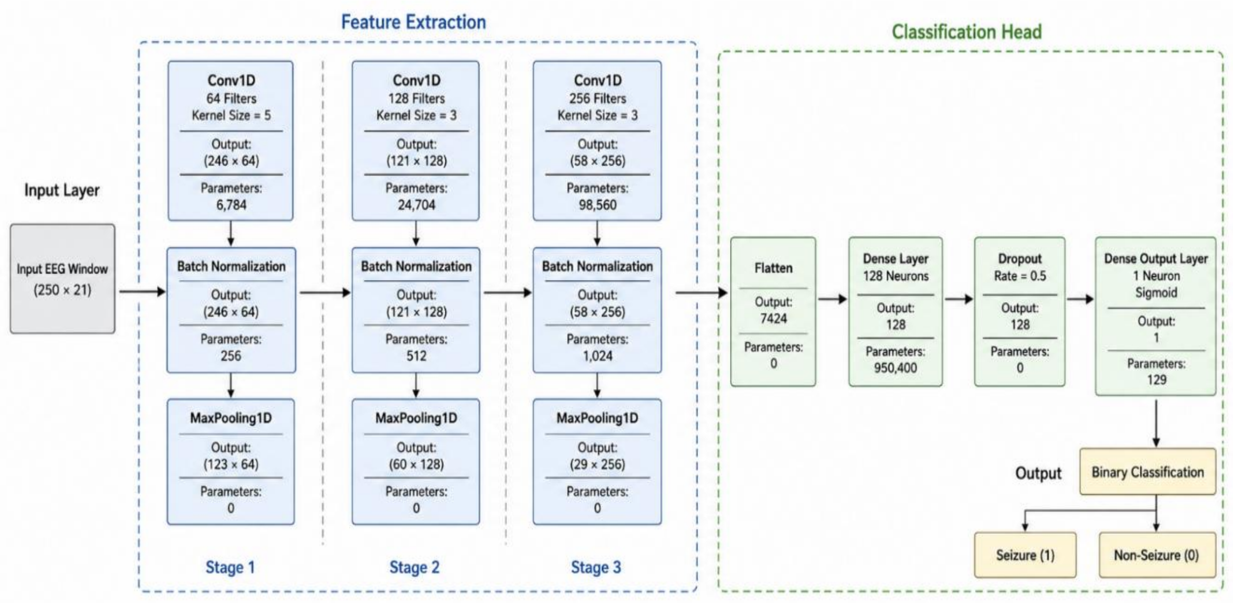

Figure 7: Proposed Conv1D convolutional neural network (CNN) architecture

Layer	Output Shape	Parameters
Conv1D (64, k=5)	(246, 64)	6,784
Batch Normalization	(246, 64)	256
MaxPooling1D	(123, 64)	0
Conv1D (128, k=3)	(121, 128)	24,704
Batch Normalization	(121, 128)	512
MaxPooling1D	(60, 128)	0
Conv1D (256, k=3)	(58, 256)	98,560
Batch Normalization	(58, 256)	1,024
MaxPooling1D	(29, 256)	0
Flatten	7424	0
Dense (128)	128	950,400
Dropout (0.5)	128	0
Dense (1)	1	129

Table 6: Applied Conv1D CNN Architecture Parameters

3.3.3 Model Compilation and Optimization

After constructing the proposed Conv1D architecture, the model was compiled and optimized using the TensorFlow and Keras deep learning framework. The Adam optimization algorithm was selected due to its computational efficiency and strong performance in deep learning applications involving large-scale biomedical time-series data. Adam combines adaptive learning rate optimization with momentum-based gradient updates, enabling faster and more stable convergence during training.

The model was trained using the binary cross-entropy loss function, which is commonly used for binary classification tasks involving probabilistic outputs. Since the proposed framework focused on distinguishing seizure and non-seizure EEG activity, the output layer generated seizure probability values ranging between 0 and 1 using sigmoid activation. Binary cross-entropy was therefore suitable for measuring the difference between predicted seizure probabilities and the corresponding ground-truth labels.

An initial learning rate of 0.001 was used during optimization to balance convergence speed and training stability. Multiple evaluation metrics were monitored throughout training, including classification accuracy, precision, recall, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). While accuracy provided a general measure of classification performance, additional emphasis was placed on recall and AUC due to the severe class imbalance present within the EEG dataset. In seizure detection applications, recall is particularly important because it measures the model's ability to correctly identify seizure events and reduce missed seizure detections.

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Cross-Entropy
Output Activation	Sigmoid
Input Shape	(250, 21)
Metrics	Accuracy, Precision, Recall, AUC

Table 7: Applied Conv1D CNN Hyperparameters and Compilation

3.4 Training and Testing

3.4.1 Class Imbalance Handling

One of the primary challenges encountered during the development of the seizure classification model was the highly imbalanced distribution of EEG windows. Following preprocessing and segmentation, non-seizure windows represented most of the training samples, whereas seizure windows constituted only a small fraction of the dataset. Such imbalance can bias the learning process toward the majority class, allowing the model to achieve high classification accuracy while failing to correctly identify clinically important seizure events.

To mitigate this issue, a class-weighting strategy was incorporated during model training. The weight assigned to each class was calculated inversely proportional to its frequency within the training dataset according to the following equation

$$W_C = \frac{N}{K \times N_C}$$

where (W_C) denotes the weight assigned to class (c), (N) represents the total number of training samples, (K) is the total number of classes, and (N_C) is the number of samples belonging to class (c). As a result, the minority seizure class received a larger weight than the majority non-seizure class, increasing its contribution during model optimization.

The computed class weights were subsequently integrated into the binary cross-entropy loss function to penalize seizure misclassification more heavily than non-seizure errors. The weighted loss function can be expressed as:

$$\mathrm{L} = -\frac{1}{N}\sum_{i=1}^{N}[w_{yi} \cdot (y_i \ln(\hat{y}_i) + (1 - y_i) \ln(1 - \hat{y}_i))]$$

where (L) denotes the weighted loss value, (w_{yi}) represents the class weight associated with sample (i), (y_i) is the true binary label, and ($\hat{y}_i$) is the predicted seizure probability generated by the model. This weighting mechanism encouraged the optimizer to place greater emphasis on correctly learning seizure-related patterns rather than minimizing errors primarily on the dominant non-seizure class.

Unlike oversampling and under sampling approaches, the adopted weighting strategy preserved the original distribution and temporal characteristics of the EEG recordings while still compensating for class imbalance during optimization. This approach was particularly advantageous for EEG analysis, as artificial duplication or removal of windows may distort the natural characteristics of seizure activity and introduce unwanted bias into the learning process.

In addition to class-weighted optimization, model performance was assessed using multiple evaluation metrics beyond overall accuracy. Since accuracy alone can provide a misleading indication of performance in highly imbalanced datasets, particular emphasis was placed on precision, recall, F1-score, specificity, and ROC-AUC.

Class	Number of Windows	Percentage
Non-Seizure (0)	323,479	94.79%
Seizure (1)	17,774	5.21%
Total	341,253	100%

Table 8: Distribution of Labels per Windows for the Scaled Subset

Class	Number of Windows	Percentage
Non-Seizure (0)	3,101,982	95.20%
Seizure (1)	156,317	4.80%
Total	**3,258,299**	**100%**

Table 9: Distribution of Labels per Windows for the Full Dataset

The scaled training dataset contained 323,479 non-seizure windows and 17,774 seizure windows, corresponding to an approximate imbalance ratio of 18.2:1. This substantial disparity demonstrates the inherent challenge of seizure detection and highlights the necessity of incorporating imbalance-aware training strategies.

Class	Weight
Non-Seizure (0)	0.527
Seizure (1)	9.600

Table 10: Class Weights for the Scaled Subset

Class	Weight
Non-Seizure (0)	0.524
Seizure (1)	10.994

Table 11: Class Weights for the Full Dataset

Applying the weighting scheme to the training subset resulted in a weight of 0.527 for the non-seizure class and 9.600 for the seizure class. Consequently, seizure classification errors contributed approximately 18 times more to the optimization process than non-seizure errors, encouraging the model to focus on the minority seizure class despite its limited representation within the training data.

Several alternative imbalance mitigation techniques were reviewed during the system design phase, including random oversampling, random undersampling, synthetic sample generation methods such as SMOTE, and specialized loss functions including focal loss. While these approaches have demonstrated effectiveness in various imbalanced classification problems, they were not implemented within the binary seizure detection framework of this study.

Random oversampling may increase the risk of overfitting through repeated seizure samples, whereas undersampling can discard valuable non-seizure EEG information. Synthetic data generation techniques may also alter the physiological characteristics of EEG signals and introduce artificial patterns. Consequently, class-weighted optimization was selected as the primary imbalance handling strategy because it preserved the original EEG data distribution while compensating for class imbalance directly during model training.

3.4.2 Training Callbacks and Regularization

To improve model generalization and reduce overfitting, several regularization techniques and training control mechanisms were incorporated into the Conv1D classification framework. Since deep learning models contain many trainable parameters, they may learn noise and dataset-specific patterns rather than generalized seizure-related EEG characteristics. Therefore, multiple strategies were employed to improve training stability and model robustness.

Batch normalization layers were inserted after each convolutional layer to normalize intermediate feature distributions during training. This technique improves gradient flow, accelerates convergence, and stabilizes the optimization process by reducing variations in feature magnitudes across mini batches.

A dropout layer with a dropout rate of 0.5 was applied before the final classification layer. During training, dropouts randomly deactivate a portion of neurons, preventing excessive dependence on specific network connections and reducing the risk of overfitting. This encourages the model to learn more generalized EEG representations that perform better on unseen recordings.

Several training callbacks were also implemented to monitor validation performance and automatically control the optimization process. Early stopping was used to terminate training when validation loss failed to improve for a predefined number of epochs, preventing unnecessary training and reducing overfitting. In addition, a learning rate scheduler based on the ReduceLROnPlateau strategy was employed to decrease the learning rate when validation performance reached a plateau, allowing finer parameter updates during later training stages.

Finally, a model checkpointing mechanism was incorporated to automatically save the best-performing model according to validation loss. This ensured that the final evaluation was performed using the most generalized model obtained during training rather than simply using the final training epoch.

Parameter	Value
Optimizer	Adam
Initial Learning Rate	0.001
Batch Size	256
Maximum Epochs	50
Batch Normalization	Applied after each Conv1D layer
Dropout Rate	0.5
Early Stopping	Enabled
ReduceLROnPlateau	Enabled
Learning Rate Reduction Factor	0.5
Model Checkpointing	Enabled
Training Data Shuffling	Enabled

Table 12: Checkpointing Mechanisms Incorporated in the Training

3.4.3 Training Pipeline

Training deep learning models on large-scale EEG datasets presents significant computational challenges due to the high dimensionality and volume of EEG recordings. Following preprocessing and window segmentation, the generated dataset contained hundreds of thousands of EEG windows, making it impractical to load the entire dataset into memory simultaneously. Consequently, a scalable training pipeline was developed to enable efficient data handling and future deployment of high-performance computing resources.

A custom data generator was implemented to support batch-wise loading of EEG windows directly from disk. The generator continuously streamed mini batches of data into memory, allowing the model to train on datasets significantly larger than the available capacity. In addition, data shuffling was performed during training to improve sample diversity within batches and reduce the risk of learning order-dependent patterns.

3.4.4 Model Evaluation Strategy

Following completion of the training phase, the performance of the proposed Conv1D seizure classification model was evaluated using an independent evaluation dataset that remained completely unseen during training and validation. The evaluation set was used exclusively for final performance assessment to ensure an unbiased estimation of the model's generalization capability on new EEG recordings.

Due to the highly imbalanced nature of seizure detection datasets, model evaluation was not based solely on overall accuracy. Although accuracy provides a general indication of classification performance, it may produce misleading results when the majority class significantly outnumbers the minority class. Consequently, multiple complementary evaluation metrics were employed to assess different aspects of model behavior.

Precision was used to measure the proportion of predicted seizure windows that corresponded to actual seizure events, while recall (sensitivity) quantified the model's ability to correctly identify true seizure occurrences. The F1-score was utilized as a balanced measure that combines both precision and recall into a single performance indicator. In addition, specificity was calculated to evaluate the model's ability to correctly identify non-seizure EEG activity and minimize false alarm generation.

To further assess the discriminative capability of the classifier across different decision thresholds, Receiver Operating Characteristic (ROC) analysis was performed and summarized using the Area Under the Curve (AUC) metric. The ROC-AUC score provides a threshold-independent measure of classification performance and reflects the model's ability to distinguish between seizure and non-seizure EEG windows.

Furthermore, confusion matrix was generated to provide a detailed visualization of classification outcomes, including true positives, true negatives, false positives, and false negatives. This analysis enabled a comprehensive assessment of seizure detection performance and facilitated the identification of strengths and limitations within the proposed classification framework.

The combination of these evaluation metrics provided a comprehensive performance assessment from both machine learning and clinical perspectives, ensuring that seizure detection capability was evaluated beyond simple classification accuracy.

To provide a quantitative assessment of classification performance, the evaluation metrics used in this study are defined as follows.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP+FN}$$

$$\text{Specificity} = \frac{TN}{TN+FP}$$

$$\text{F1 score} = 2 \times \frac{Precision \times Recall}{Precesion + Recall}$$

Where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

3.5 Full Data-Set Implementation

Full-dataset training was conducted on an AI Lab high-performance workstation equipped with 188 GB RAM and dual NVIDIA Quadro RTX 5000 GPUs (16 GB VRAM each). This hardware configuration provided sufficient computational and storage resources for large-scale EEG preprocessing, dataset generation, and model training on the full size data set.

Chapter 4 RESULTS AND DISCUSSION

4.1 Evaluation Framework and Baseline Performance

Following the execution of the multi-stage system preprocessing pipeline, the proposed 1-Dimensional Convolutional Neural Network (Conv1D CNN) architecture was deployed for final validation. To rigorously evaluate the model's real-world clinical viability, evaluation was conducted on the full, patient-disjoint test stream of the Temple University Hospital (TUH) EEG Seizure Corpus (Version 2.0.3).

Ensuring that the evaluation dataset contained no overlapping data or sessions from patients included in the training or development sets was vital to accurately test patient-independent generalization. Raw EEG data varies significantly across demographics, electrode contact impedances, and individual neurophysiological baseline profiles, making this independent validation a true test of the model's robustness.

In clinical electroencephalography, evaluating automated diagnostics solely through crude global metrics like overall classification accuracy introduces significant medical risk. Due to the sparse and transient nature of epileptic events in long-term ICU or ambulatory monitoring, non-seizure background activity naturally dominates the data. A naive or broken classifier that uniformly labels all incoming streams as background activity would mathematically demonstrate a high accuracy score while failing completely to flag critical clinical events.

To prevent this bias and provide a transparent look at system performance, this project utilizes a multi-dimensional matrix of validation metrics. This framework incorporates Overall Accuracy, Specificity, Sensitivity or Recall, Precision, and the F1-Score, which serves as the harmonic mean balancing precision and recall.

Metric	Baseline Value
Overall Accuracy	92.01%
Sensitivity (Recall)	74.15%
Specificity	93.82%
Precision	54.97%
F1-Score	0.6313

Table 13: Performance Metrics of the Full Dataset

To assess the model's discriminative capabilities independently of any single arbitrary decision threshold, a Receiver Operating Characteristic (ROC) analysis was conducted over the complete test stream. As illustrated in the figure below, the system achieved an exceptional Full Baseline ROC Area Under the Curve (AUC) of 0.972.

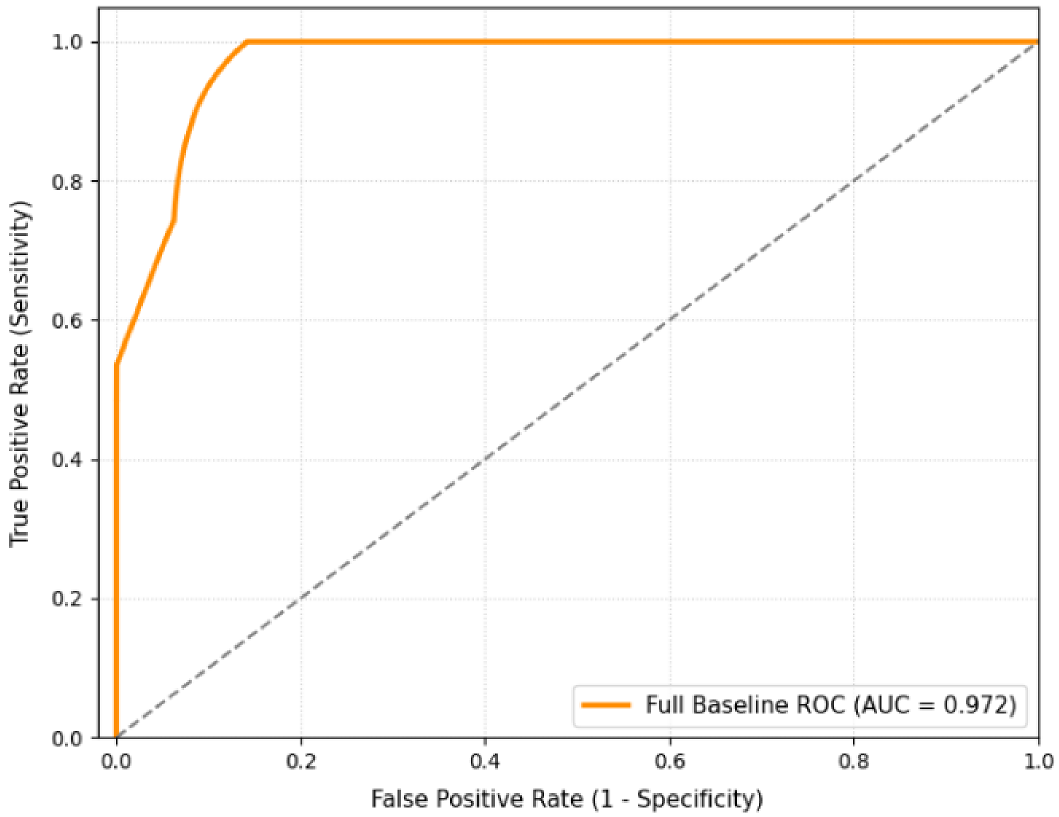

Figure 8: ROC Area Under the Curve for the Full Dataset Training

The geometric trajectory of the orange ROC curve shows a sharp, nearly vertical ascent in the early False Positive Rate (FPR) regions, maintaining a high True Positive Rate (TPR) even at strict operational thresholds. This threshold-independent performance proves that the combination of standardizing montages, applying digital bandpass filtering (0.5 Hz to 40 Hz), and using hierarchical 1D convolutional feature extraction allows the network to effectively separate real epileptic discharges from complex background brain rhythms.

4.2 Class-Specific Performance Breakdown

A granular investigation of individual class metrics explains how the class-weighting optimization scheme altered the network's internal loss landscape. In highly imbalanced datasets, unweighted empirical risk minimization causes deep networks to align their gradients with the majority class to quickly minimize loss, ignoring the critical features of the minority class.

As shown in the final classification report below, the evaluation mirrors these real-world clinical challenges, presenting a massive pool of 510,875 independent, 1-second temporal windows. The distribution across the two target conditions is heavily skewed:

- Non-Seizure (BCKG) Class Support: 463,721 windows
- Seizure Class Support: 47,154 windows

This creates an inherent class imbalance ratio of approximately 9.83:1 in the test stream. Under these strict testing conditions, the class-specific performance is broken down into its operational components:

For the dominant background class, the Conv1D network achieved a precision of 0.9727, a recall of 0.9382, and a final F1-score of 0.9552. The high precision value is clinically significant; it ensures that when the system classifies a sequence of windows as normal background activity, there is a very high probability that the patient is stable. This reliability reduces the need for medical staff to manually review large volumes of normal baseline data.

Metric	Baseline Value
Precision	0.9727
Recall	0.9382
F1-Score	0.9552

Table 14: Performance Metrics for the Dominant Background Class

For the critical minority seizure class, the model achieved a precision of 0.5497, a recall of 0.7415, and an F1-score of 0.6313. While a standard unweighted neural network often fails under a 9.8:1 imbalance, frequently yielding zero or near-zero seizure recall, the incorporated inverse-frequency class weighting scheme successfully adjusted the model's decision boundaries.

During the model compilation stage (detailed in Chapter 3), a class penalty weight of 10.994 was assigned to the seizure class, while the background class was weighted at 0.524. When integrated directly into the binary cross-entropy loss function, this weighting scheme multiplied the loss incurred by any missed seizure window (FN) or false seizure alarm (FP) by a factor of nearly 21 relative to background classification errors. This adjustment forced the Adam optimizer to prioritize learning the localized temporal patterns of epileptic events, such as paroxysmal spike-and-wave discharges, polyspike complexes, and rhythmic delta-theta runs.

Metric	Baseline Value
Precision	0.5497
Recall	0.7415
F1-Score	0.6313

Table 15: Performance Metrics for the Minority Seizure Class

4.3 Confusion Matrix and Error Analysis

4.3.1 Confusion Matrix

The baseline normalized confusion matrix presented maps the system's operational trade-offs, providing an exact percentage breakdown of the network's classification decisions across nearly half a million validation windows:

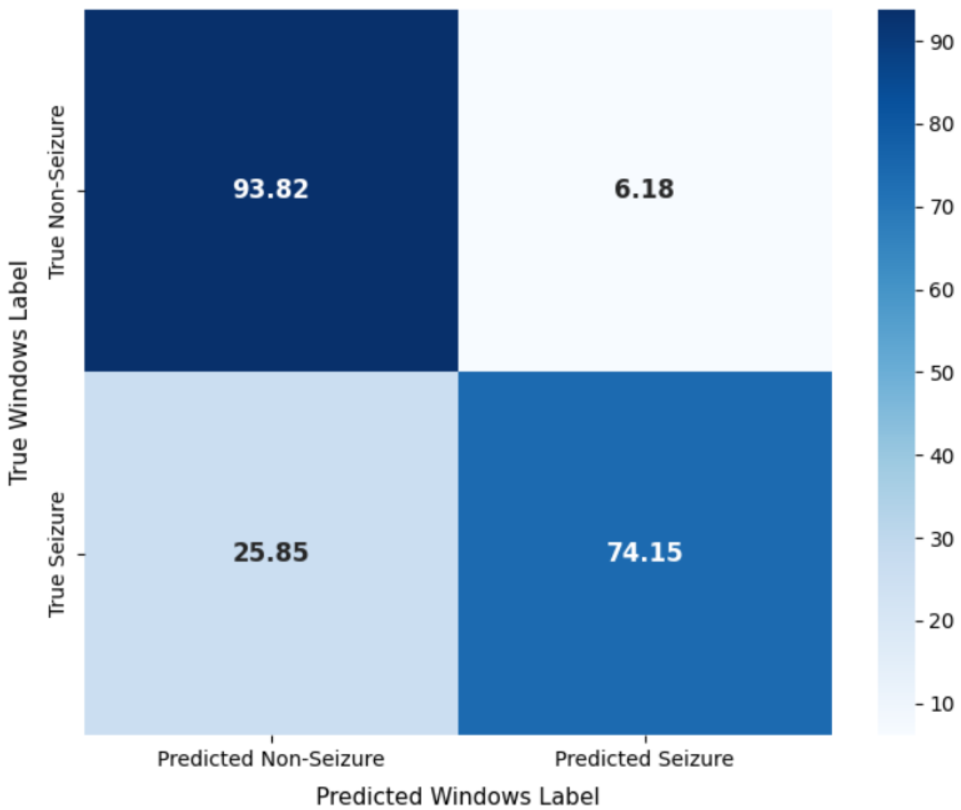

Figure 9: Full Dataset Confusion Matrix

- True Negatives (93.82%): Out of 463,721 actual non-seizure windows, the network correctly classified 435,063 segments as baseline background activity, demonstrating high stability over long monitoring periods.

- False Positives (6.18%): The network incorrectly flagged 6.18% of normal background activity as an active epileptic seizure. In a clinical hospital environment, keeping the false positive rate low is essential to prevent "alarm fatigue," an issue where nursing staff begin to ignore automated alerts due to constant false alarms from overly sensitive monitoring equipment.

- True Positives (74.15%): The system successfully detected 74.15% of all active seizure windows, proving that a deep learning model can automatically capture meaningful epileptic patterns across an unconstrained multi-patient clinical stream.

- False Negatives (25.85%): The network missed 25.85% of active seizure windows, incorrectly categorizing them as normal background brain activity.

4.3.2 Error Analysis

To improve future iterations of this system, it is vital to analyze the underlying causes of the 25.85% false negative rate and the 54.97% precision score:

Electromyographic (EMG) and Electrooculographic (EOG) Artifacts: Scalp-mounted EEG electrodes pick up any electrical charge generated near the cranium. As a result, physiological artifacts like muscle clenching (EMG), rapid eye movements or blinking (EOG), swallowing, and patient motion introduce large, high-frequency voltage spikes into the signal. These artifacts can distort or completely mask the underlying neurophysiological seizure patterns. Because these artifacts often appear as sharp, rhythmic transients, they can confuse the Conv1D layers, leading the model to generate false positives that lower overall precision.

Electrographic Polymorphism Across Seizure Subtypes: As detailed in the dataset description in Chapter 2, the merged seizure class contains a highly diverse mix of seizure morphologies (e.g., Focal Non-Specific Seizures [FNSZ], Generalized Seizures [GNSZ], Complex Partial Seizures [CPSZ], and Absence Seizures [ABSZ]). A generalized tonic-clonic seizure produces high-amplitude, synchronous rhythmic runs across almost all 21 channels, which the model can easily detect. In contrast, certain complex partial or focal non-specific seizures produce subtle, highly localized frequency shifts or low-voltage flattening restricted to a few electrodes. Because the network processes all channel uniformly, these subtle localized patterns can be washed out by the surrounding normal activity on other channels, leading to false negatives.

4.4 Technical Justification of Performance Gaps

The empirical validation of the proposed Conv1D architecture yields a highly polarized performance profile. To transition this system from a statistical output into a meaningful biomedical engineering analysis, the underlying mechanisms drive both the system's operational strengths and its vulnerabilities must be thoroughly justified.

4.4.1 The Strengths: Justifying the High Global Metrics and Discriminative Depth

1. Robust Temporal Translation Invariance via Convolutional Weights

The model's high global accuracy and outstanding threshold-independent classification capability are technically justified by the architectural compatibility of 1D convolutional layers with non-stationary time-series data.

Unlike traditional multi-layer perceptrons or machine learning pipelines that treat temporal samples as independent variables, the sliding kernels of a Conv1D layer enforce an inductive bias of temporal translation invariance.

In clinical EEGs, a paroxysmal spike-and-wave discharge or a rhythmic delta run can manifest at any arbitrary millisecond within a segmented 1-second window depending on when the physical seizure focus discharges. Because the convolutional kernels share weights across the entire temporal sequence, the network isolates these high-amplitude, rapidly evolving wave morphologies regardless of their exact temporal onset within the block.

2. Hierarchical Feature Extraction Layering

The progression from a broad kernel size (k=5) in the initial layer to narrower kernels (k=3) in subsequent blocks mathematically optimizes feature discovery:

- Layer 1 (Low-level primitives): The k=5 kernel captures broad, low-frequency phase shifts and macroeconomic voltage transitions across the 0.5Hz to 40 Hz passband.
- Layers 2 & 3 (High-level abstract dynamics): The narrower k=3 kernels extract micro-temporal patterns, such as the sharp rise times of spikes or the phase transitions occurring between a tonic discharge and a clonic burst.

This hierarchical extraction explains why the model maintains an exceptionally true positive rate across a broad spectrum of decision thresholds, as visualized by the sharp vertical ascent of the ROC curve.

4.4.2 The Weaknesses: Justifying the Performance Gaps and Trade-Offs

1. Mathematical Bias of Asymmetric Gradient Scaling (Recall vs. Precision)

The prominent operational gap between the minority class sensitivity and its positive predictive value is a direct mathematical consequence of altering the network's optimization landscape via extreme class-weight penalties.

When integrated into the binary cross-entropy loss equation, this massive disproportion scales the gradients during backpropagation:

$$W \propto W_1 \cdot (y_i - \hat{y_i})$$

Every time the model encounters an ambiguous, noisy, or transitional boundary window and incorrectly predicts a background state (a False Negative), the weight updates are penalized approximately 21 times harder than if it misclassified a background window (a False Positive).

Consequently, the Adam optimizer forces the internal weights of the fully connected layers to retreat to a highly defensive decision boundary. The network minimizes its aggregate loss by systematically lowering its probabilistic activation threshold for the seizure class.

While this successfully protects the clinical requirement of high sensitivity, ensuring that transient or low-amplitude seizure windows are aggressively captured, it inherently forces the model to misclassify borderline noisy background frames. This optimization bias mathematically justifies the high volume of false alarms and the resulting drop in precision.

2. Spatial Blindness of Flattened Temporal Uniformity (False Negatives)

The 25.85% false negative rate represents clinical intervals where active seizures were entirely missed by the network. This failure mode is structurally justified by the spatial blindness of standard 1D CNN architectures.

The model reads the multichannel EEG input as a stacked matrix where the 21 electrodes are compressed into uniform feature channels. When the network executes a Flatten operation to transition from convolutional blocks to dense layers, it treats the spatial topology of the international 10–20 scalp placement system as a flat, Euclidean vector.

- The Clinical Failure Mode: While generalized seizures involve synchronous, high-amplitude discharges across all 21 electrodes that easily saturate the network’s dense layers, focal or complex partial seizures (FNSZ or CPSZ) are highly localized. They often manifest micro-volt localized changes confined to a narrow anatomical region.

- The Numerical Washout: Because the 1D temporal kernels process all channels uniformly without modeling their functional connectivity or physical distance on the scalp, the normal background electrical activity happening simultaneously across the remaining 18 unaffected channels mathematically drowns out the localized, low-amplitude abnormal features of the focal seizure. The spatial topology is lost, causing the network to misclassify these localized events as baseline background activity.

3. Morphological Mimicry of Unfiltered Physiological Artifacts (False Positives)

The low precision score (54.97%) highlights a vulnerability to non-cerebral electrical potentials. Because scalp-mounted electrodes read signals through the skin, they record any electrical activity generated within the cranium, creating high-frequency artifact patterns that match true seizure characteristics:

Artifact Source	Physical Cause	Morphological Characteristic	Network Misinterpretation
Electromyographic (EMG)	Jaw clenching, swallowing, head movement	Sharp, high-frequency, highly rhythmic voltage bursts	Interpreted by Conv1D kernels as paroxysmal poly-spike complexes or active tonic runs.
Electrooculographic (EOG)	Eye blinking, saccadic eye tracking	High-amplitude, low-frequency localized deflections	Passed by the 0.5Hz filter and interpreted as rhythmic delta activity or slow-wave seizure discharges.

Table 16: Possible Network Misinterpretation of Physiological Artifacts

CHAPTER 5 SYSTEM INTEGRATION AND SOFTWARE ARCHITECTURE

5.1 Overview of system architecture

5.1.1 Introduction

The "AI-powered cloud-based healthcare system" is architected as a decoupled, multi-tier web application designed for high availability, security, and low-latency inference. The integration bridges a lightweight client-facing User Interface (UI) with a compute-optimized Amazon Web Services (AWS) Elastic Compute Cloud (EC2) backend harboring a deep learning pipeline.

The system architecture is structured into three distinct operational layers:

1. Presentation Layer (Frontend UI): A client-side web interface engineered using HTML5, CSS3, and JavaScript. It operates asynchronously to capture clinical metadata and ingest data files without blocking the user experience.

2. Application & Inference Layer (Backend): An AWS EC2 Ubuntu instance configured with a virtual Python environment. It hosts a lightweight web server framework (such as Flask) alongside the initialized deep learning dependencies ($TensorFlow$, $NumPy$, and $SciPy$) to orchestrate execution flows.

3. Data Management Layer: Controlled via local structured directory endpoints on an attached Amazon Elastic Block Store (EBS) volume, managing incoming raw evaluation files and model weight repositories.

5.1.2 Architectural Paradigm: Embracing MLOps

The engineering workflow executed in this project transcends traditional software deployment by strictly adhering to the principles of MLOps (Machine Learning Operations). MLOps represents the intersection of Machine Learning, DevOps (Development Operations), and Cloud Computing, fundamentally aimed at unifying model development with automated, reliable production deployment. Traditional DevOps focuses heavily on static code execution and continuous integration of standard software builds. In contrast, the integration of this EEG seizure detection system requires a specialized MLOps paradigm because it must simultaneously manage code variations, data pipeline state transitions, and live model artifact behavior.

This project aligns directly with the core pillars of MLOps through three distinct lifecycle phases:

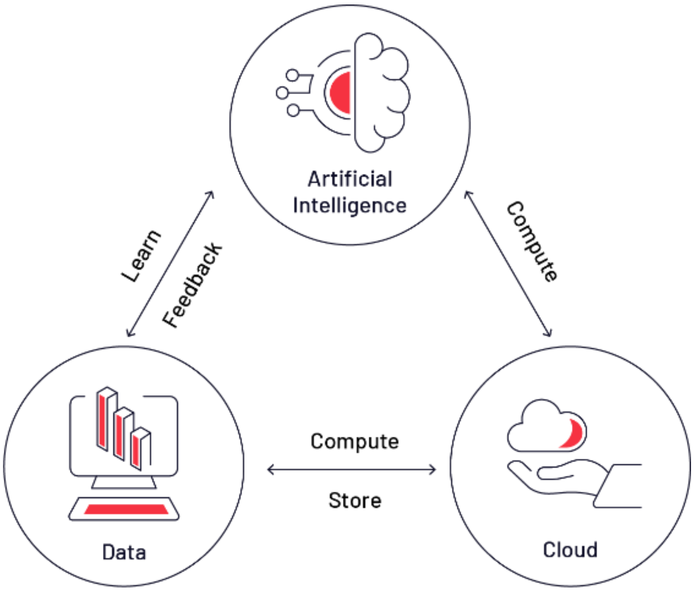

Figure 10 Integrating Machine Learning with Cloud Deployment

- Machine Learning & Inference Engineering: Rather than focusing on model training variables, the deployment pipeline isolates a pre-trained, static Convolutional Neural Network (CNN) weight file. By loading the model into server memory a single time during application startup, the pipeline optimizes computational efficiency and avoids repeated initialization overhead.
- DevOps Best Practices: The system decouples the front-end presentation layer from the analytical engine to ensure modular maintenance. Furthermore, to counteract common environment drift and dependency collisions when transitioning from local development to production, the backend is strictly bound to locked framework dependencies within an isolated virtual project environment.
- Cloud Deployment & Asset Management: The architecture utilizes compute-optimized AWS EC2 infrastructure combined with persistent Elastic Block Store (EBS) volumes. System reliability is reinforced by enforcing explicit absolute paths and rigid working directory controls to manage the ingestion of incoming (.Npz) data file segments.

5.2 System Design

5.2.1 UI Component Division

The user interface divides the screen real estate into two highly functional, high-contrast panels to separate input operations from diagnostic outputs:

- Patient Information Panel (Ingestion Node): Positioned on the left, this panel contains fields for the Patient Name, Phone Number, an .npz file upload drop-zone, and an execution trigger labeled "Analysis".
- Analysis Result Panel (Diagnostic Output Node): Positioned on the right, this panel acts as a deterministic medical alert mechanism that updates dynamically upon backend response delivery, avoiding overly complex charts to maximize rapid clinical interpretation with the suitable medical recommendations for the occurred seizure.

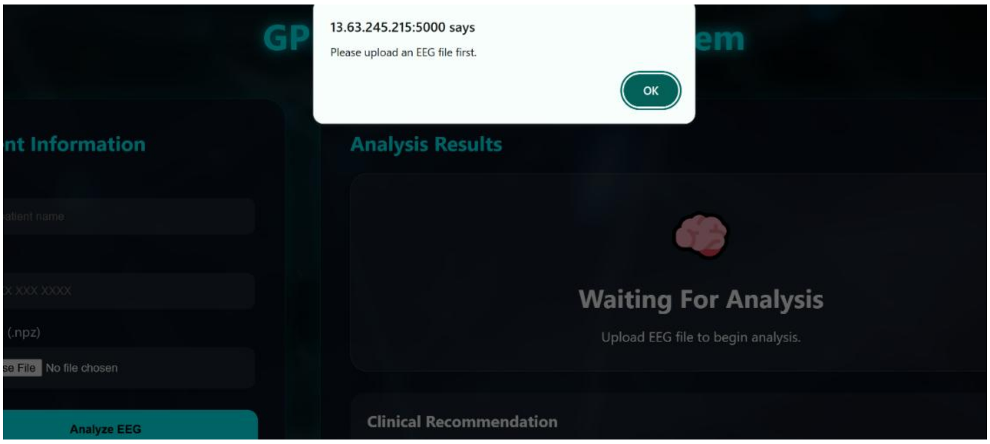

Figure 11 UI when no EEG file or data is uploaded

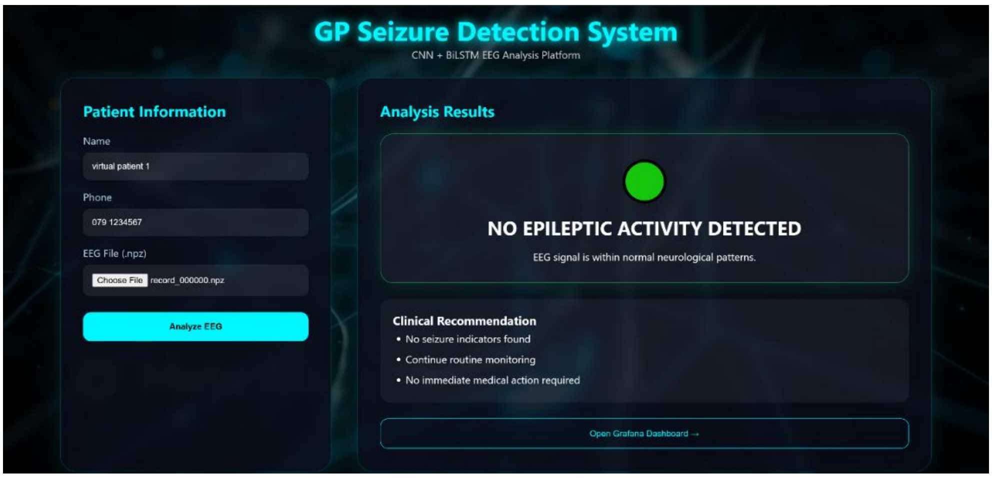

Figure 12 UI when the patient is seizure-free

Figure 13 UI when seizure is detected and the alarm is triggered

5.2.2 Design Philosophy and Accessibility

To ensure usability under varying clinical environments, the interface implements specific front-end design techniques:

- Glassmorphism Panels: Utilizes semi-transparent structural containers with background blur effects to balance readability against an underlying neural-themed animated video backdrop.
- Strict UX Color Coding: Adheres to established medical monitoring conventions to guide visual triage:

Interface Element / State	**Design Color**	**Clinical Meaning / Context**
System Operations & Navigation	Cyan	Standard system functionality, neutral state.
Normal EEG Activity	Green	Negative screening; normal activity, low-risk styling.
Interface Element / state	Design color	Clinical Meaning/context
Potential Seizure Activity	Red	Positive screening, abnormal patterns, emergency, warning.

Table 17 Design Specifications

5.3 Memory Management and Initialization Strategy

To fulfill clinical efficiency standards, the system separates the training pipeline from the cloud runtime deployment. The CNN model is loaded into server memory exactly once via TensorFlow's Keras API during backend application startup.

Integration Safeguard: Initializing the model globally at startup completely eliminates repeated file I/O reloading overhead during sequential user upload requests, ensuring minimal inference response times .

5.4 Data Ingestion Contract

The system mandates a standardized data exchange format to guarantee integration stability:

- File Format: Input data must be structured as pre-segmented numerical archives in .npz format.
- Data Conformance: The arrays within the .npz file must strictly match the deterministic input shape tensor expected by the compiled Keras model layer architecture.

5.5 End-to-End Integration Workflow

The system establishes a tightly coupled, sequential runtime loop across its multi-tier infrastructure when processing a file:

- **Ingestion Execution:** The clinician fills in patient identifiers, links the .npz data file, and triggers the analysis action.
- **Payload Transmission:** The client-side application packages the fields and streams the file payload asynchronously over HTTP to the AWS EC2 hosted server endpoint.
- **Context Definition:** The backend web server captures the stream and isolates execution within an explicitly enforced absolute path workspace to avoid directory routing faults.
- **Window-Based Processing Deconstruction:** The raw EEG signal is split sequentially into discrete, fixed-length sliding temporal windows (W_t) to isolate localized seizure anomalies rather than broad global signal averaging.
- **Sequential Batch Prediction:** The inference engine goes through these recordings, mapping each segment to a probabilistic score outputting class labels:

Class 0: Normal EEG Activity (Displays green status, low-risk messaging, standard clinical recommendations).

Class 1: Epileptic Seizure Activity (Triggers red warning indicator, medical alert, emergency prioritization directives).

- **State-Driven Presentation Render:** The backend generates a JSON response payload containing the classified outcome, transmitting it back to the client browser to immediately update the visual styling of the Analysis Result Panel.

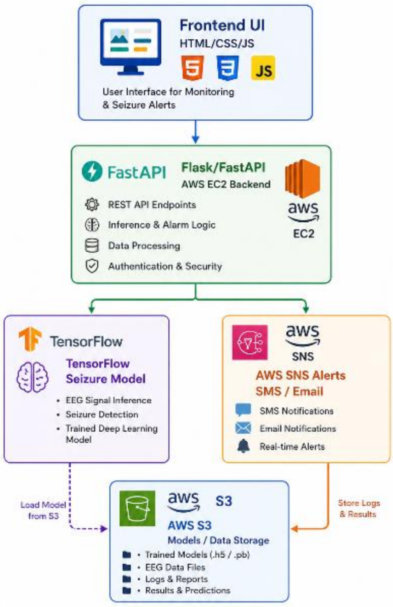

Figure 14 System Architecture and data flow through the cloud-server

5.6 Technical Challenges and Architectural Solutions

During the integration and migration of the system from a localized testbed to the AWS EC2 cloud instance, several engineering roadblocks were encountered and resolved:

5.6.1 Environment Path Discrepancies

- **Problem:** Relative file paths caused immediate runtime failures (FileNotFoundError) upon server execution due to varying shell context configurations on the Ubuntu server.
- **Solution:** Switched the data pipeline from relative addressing to strict, absolute paths, hard-coding a dedicated working directory configuration at the root initialization file of the pipeline.

5.6.2 Missing Remote Dependencies

- **Problem:** Environment collisions and missing system libraries caused core processing crashes during the deployment phase.
- **Solution:** Encapsulated the application dependencies within a dedicated Python virtual environment, locking specific framework versioning for TensorFlow, NumPy and imaging libraries to guarantee consistent environment state replication.

5.6.3 CPU Computational Latency Constraints

- **Problem:** High processing latency occurred when running heavy sequential sliding window loops across massive EEG arrays on standard compute-restricted CPU EC2 configurations.
- **Solution:** Refactored the inference engine loop to minimize redundant data transformation copies in memory, implemented batch-wise parallel execution transformations, and restricted the model instantiation strictly to system startup.

5.7 Production Scalability and Future Roadmap

To transition this validated architectural prototype into an enterprise healthcare framework serving multiple institutions, a multi-phased scalability roadmap has been strategically drafted:

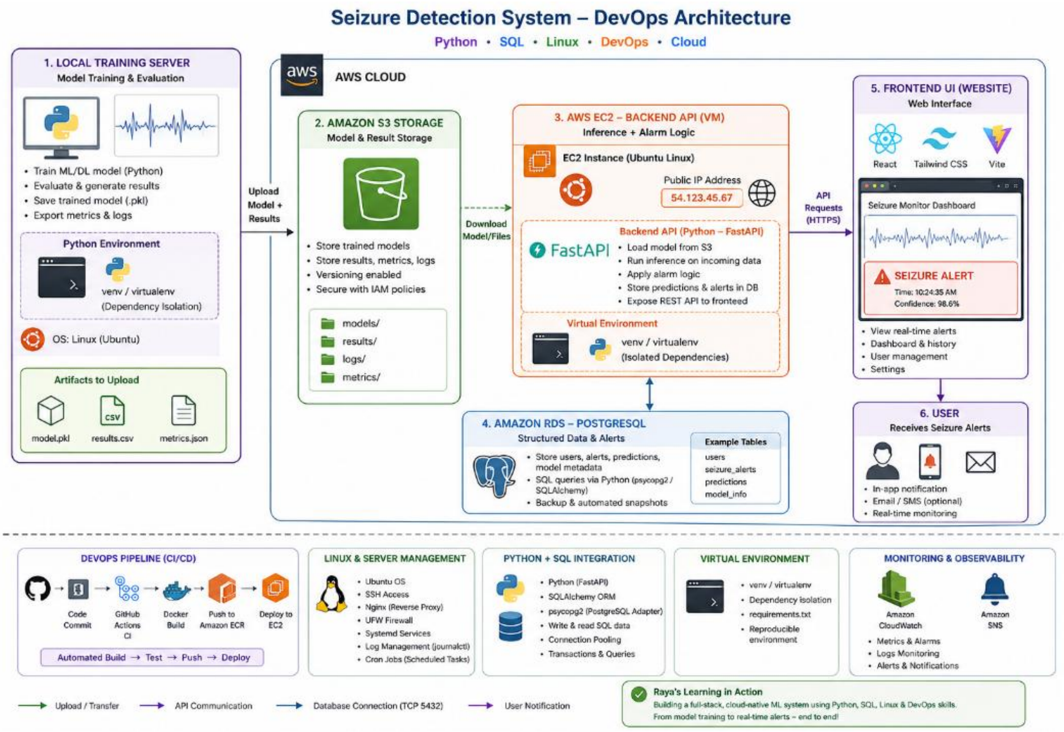

Figure 15 Future improvements roadmap

5.7.1 Containerization and Orchestration (Infrastructure Tier)

- **Docker Isolation:** Packaging application layers into distinct microservice containers (Web UI container, Machine Learning Inference engine container, Database container) ensures uniform execution states.
- **Kubernetes Orchestration:** Deploying containers within a Kubernetes cluster introduces automated health checks, self-healing node restorations, and load distribution across server topologies.

5.7.2 Enterprise Monitoring and Automated Alerts (Observation Tier)

- **Grafana Dashboards:** Integrating an analytical visualization platform hooked directly to operational data logs will allow clinical directors to monitor aggregated system performance metrics, prediction volumes, and confidence distribution scores.
- **SMS & Email Communications:** Utilizing third-party communication layers (e.g., AWS SNS, Twilio) to dispatch immediate, automated alert text notifications and comprehensive PDF medical files to registered caregivers right after a positive seizure detection event.
- **Database-Driven Records Management:** Replacing local storage with a dedicated PostgreSQL or MySQL relational layer to capture patient logs, processing history, and tracking long-term clinical trends.

5.7.3 Advanced Processing and Autonomous Scaling (Core Tier)

- **Real-Time Data Streaming Ingestion:** Upgrading the system from batch-oriented .npz file processing to persistent web sockets or cloud streaming nodes to evaluate live, ongoing EEG feeds straight from wearable patient monitors.
- **AWS Auto Scaling Policies:** Utilizing dynamic AWS Auto Scaling groups to continuously balance compute power, expanding or shrinking instance pools dynamically based on active clinical user demands to reduce infrastructure waste ($Costs \to min$).
- **Hybrid Deep Learning Frameworks:** Exploring model retraining loops and hybrid model variations (such as CNN-LSTMs or Transformer-based architectures) to expand classification boundaries as data scale increases.

5.8 Project Summary

This thesis successfully presents the design, integration, and cloud deployment of an end-to-end, AI-powered healthcare system for the early detection of epileptic seizures using automated EEG analysis. By migrating the core machine learning infrastructure from a localized development environment to an AWS EC2 cloud instance, the platform achieves a stable, highly available runtime pipeline capable of performing automated batch or real-time inferences.

The system utilizes a trained Convolutional Neural Network (CNN) model that is loaded directly into the EC2 server memory at startup using TensorFlow's Keras API, significantly reducing computational latency by eliminating repeated I/O re-initialization overhead. The backend processing loop accepts standardized, pre-segmented numerical files in .npz format, executing a fine-grained, window-based temporal analysis to capture localized seizure anomalies rather than broad global averages. Grounded by a resilient directory management framework that implements absolute pathing, the backend reliably mitigates typical environment path discrepancies and dependency conflicts common in remote cloud instances.

Complementing this backend processing engine is a modern, responsive web-based User Interface (UI) built on standard HTML5, CSS3, and JavaScript. Adhering to professional medical dashboard standards, the front-end features a high-contrast panel design—including a Patient Information Panel for clinical data ingestion and an Analysis Result Panel for immediate diagnostic output. Through the integration of glassmorphism aesthetics, an animated neural backdrop, and strict color-coded alert triage metrics (Cyan for operation, Green for normal brain activity, and Red for seizure risks), the user interface offers healthcare operators an optimized, visually intuitive workspace for rapid clinical assessment.

5.9 Key Contributions and concluding remarks

The primary contributions realized during the development and integration of this diagnostic healthcare platform include:

- **Successful Cloud Migration:** Established a fully functional, stable machine learning inference pipeline on an AWS EC2 Ubuntu infrastructure, moving the prototype past isolated training frameworks into a deployed cloud environment.

- **Fine-Grained Temporal Inference:** Engineered a robust, window-based segmentation loop within the backend architecture to systematically process incoming .npz EEG data records, outputting probabilistic scores that align perfectly with designated binary classification thresholds.

- **Clinical UX/UI Standardization:** Designed an accessible, dual-panel medical interface that balances advanced front-end styling with practical usability, delivering direct, high-contrast medical notifications instead of complex charts to expedite patient care.
- **Resilient Environment Design:** Solved critical integration roadblocks—such as file path resolution errors and runtime environment collisions—by enforcing explicit absolute paths and isolating system dependencies within a locked Python virtual environment.

Ultimately, the developed prototype effectively bridges the gap between deep learning model architecture and practical, cloud-based application deployment. By delivering automated, reliable window-by-window analysis and translating complex mathematical inferences into instant, color-coded medical alert notifications, the system proves its viability as an impactful decision-support utility for contemporary healthcare professionals. While the current iteration functions beautifully as a standalone software pipeline, the modular nature of both its frontend layout and backend infrastructure establishes an ideal baseline for future production-grade expansions. Transforming this cloud-based architecture into a highly scaled, containerized, and real-time streaming enterprise platform will drastically improve early epileptic diagnostic workflows, maximizing patient safety and clinical efficiency on a multi-institutional scale.

CHAPTER 6 CONCLUSION AND FUTURE WORK

6.1 Project Summary and Main Findings

This graduation project successfully developed and evaluated artificial intelligence framework for the binary classification of epileptic seizures using scalp electroencephalography (EEG) signals. Validated against the highly diverse and clinically realistic Temple University Hospital (TUH) EEG Seizure Corpus (Version 2.0.3), the system demonstrates the practical viability of deep learning models in processing non-stationary biomedical time-series data within the structural constraints of an undergraduate curriculum.

The primary engineering objective, building a lightweight, computationally efficient diagnostic tool without relying on manual, handcrafted feature extraction routines, was fulfilled by designing a custom 1-Dimensional Convolutional Neural Network (Conv1D CNN). By standardizing heterogeneous multi-patient recording configurations into a unified Average Reference (AR) montage, resampling the data streams to 250 Hz, filtering out environmental and baseline noise, and generating independent 1-second temporal windows, the preprocessing pipeline effectively structured raw clinical data for deep learning applications.

When deployed on the full dataset containing over half a million evaluation windows, the system achieved a high overall accuracy of 92.01% and an exceptional threshold-independent ROC-AUC of 0.972. Because clinical seizure monitoring faces extreme class imbalances, the integration of an inverse-frequency class-weighting mechanism directly into the binary cross-entropy loss function proved vital. This strategy adjusted the network's internal gradient updates to prioritize the minority class, shifting decision boundaries to secure a strong seizure sensitivity (recall) of 74.15% and a baseline specificity of 93.82%. These metrics prove that the lightweight Conv1D architecture can capture local temporal patterns associated with epileptic discharges while maintaining a low false-positive rate, which is crucial for preventing clinical alarm fatigue.

6.2 Key Lessons Learned and Present Limitations

The development lifecycle of this mechatronics graduation project provided critical insights into handling large-scale clinical datasets and optimization challenges:

- Imbalance-Aware Optimization: Relying on global accuracy during model design can introduce significant medical risk. Incorporating a heavily penalized class-weighting scheme (weighting factor of 10.994 for seizure windows) proved to be an elegant, data-preserving solution that forced feature extraction without artificially duplicating patterns or deleting normal baseline data.

- Automated Temporal Feature Extraction Without Handcrafted Biases: The system successfully demonstrated the power of deep learning for automated feature discovery in complex biomedical signals. Unlike traditional machine learning pipelines that rely on manual feature extraction methods, the Conv1D network automatically isolated localized temporal waveform characteristics directly from the raw, noisy EEG streams. It successfully detected subtle, non-stationary epileptic signatures, such as paroxysmal spike-and-wave discharges and rhythmic delta-theta runs, achieving a high threshold-independent discriminative performance with an ROC-AUC of 0.972.

Despite these positive outcomes, several structural limitations remain within the baseline framework:

- Artifact Vulnerability: Scalp-mounted electrodes are highly sensitive to physiological artifacts such as muscle contractions (EMG) and eye blinks (EOG). Because the baseline model lacks a dedicated automated artifact rejection layer, high-frequency noise occasionally mimics epileptic transients, limiting the model's precision to 54.97%.
- Spatial Feature Blindness: While the Conv1D layers are highly optimized for localized temporal waveform patterns, they treat the 21 channels uniformly. This design misses the spatial relationships and inter-hemispheric correlations across the international 10–20 electrode placement scheme.

6.3 Future Work

To transition this baseline framework from a successful proof-of-concept into a clinically operational, deployable diagnostic tool, the following technical advancements are proposed for future development loops:

1. Transition to Hybrid Spatial-Temporal Network Architectures

To address spatial blindness and context isolation, future iterations should replace the standalone Conv1D blocks with hybrid models:

- Spatial Processing: Integrating Graph Convolutional Networks (GCNs) or Sequential Graph Convolutional Networks (SGCN) will allow the system to model the brain's non-Euclidean structural connections. By treating each electrode as a node and the anatomical distance or functional connectivity as edges, the model can isolate localized focal seizures that are restricted to specific brain hemispheres.
- Temporal Context: Appending Bidirectional Long Short-Term Memory (BLSTM) or Gated Recurrent Unit (GRU) layers after spatial feature extraction will enable the pipeline to track signal evolution across historical contexts, minimizing classification errors at transitional seizure boundaries.

2. Advanced Signal Preprocessing and Automated Artifact Rejection

Implementing automated, online artifact filtering is a priority for improving system precision. Future work will incorporate Independent Component Analysis (ICA) or Discrete Wavelet Transform (DWT) thresholding routines directly into the real-time pipeline. This addition will isolate and isolate eye-blink and muscle artifacts prior to tensor evaluation, preventing non-cerebral noise from triggering false alarms.

3. Edge-AI Hardware Deployment and IoT Integration

Reflecting the core objectives of mechatronics engineering, the compact footprint of the model (1.08 million parameters) makes it an ideal candidate for edge deployment. The next development phase will focus on:

- Model Quantization: Quantizing the trained network weights from FP32 to INT8 precision using TensorFlow Lite to minimize memory and computational overhead.
- Hardware Prototyping: Deploying the quantized network onto low-power edge microcontrollers or single-board computing modules (such as the NVIDIA Jetson Nano or Raspberry Pi) interfaced with a portable, wearable EEG headset.
- Cloud Architecture: Integrating the edge processor with an IoT Cloud Command Center via MQTT or WebSockets protocols. This framework will stream binary classification flags in real time to centralized nursing dashboards, generating immediate, automated alerts for ICU personnel during emergency status epilepticus events.

REFERENCES

1] T. S. Gill, S. S. H. Zaidi, and M. A. Shirazi, “Attention-based deep convolutional neural network for classification of generalized and focal epileptic seizures,” Epilepsy & Behavior, vol. 155, p. 109732, 2024.

[2] T. Gupta, N. Wagh, S. Rawal, B. Berry, G. Worrell, and Y. Varatharajah, “Tensor decomposition of large-scale clinical EEGs reveals interpretable patterns of brain physiology”,2022.

[3] N. Wagh, J. Wei, S. Rawal, B. Berry, L. Barnard, B. Brinkmann, G. Worrell, D. Jones, and Y. Varatharajah,“Domain-guided self-supervision of EEG data improves downstream classification performance and generalizability,” in Proceedings of Machine Learning for Health, pp. 130–142, PMLR, 2021.

[4] A. Stiehl, M. Flammer, F. Anselstetter, N. Ille, H. Bornfleth, S. Geißels¨oder, and C. Uhl, “Topological analysis of low dimensional phase space trajectories of high dimensional EEG signals for classification of interictal epileptiform discharges”, 2023.

[5] K. Jusseaume and I. Valova, “Brain age prediction/classification through recurrent deep learning with electroencephalogram recordings of seizure subjects,” Sensors, vol. 22, no. 21, p. 8112, 2022.

[6] J. Wan, S. Xu, J. Lin, L. Ji, J. Cheng, Z. Li, X. Qu, and C. Li, “Ai-enhanced wearable technology for human physiological signal detection: Challenges and future directions,” Small, p. e04078, 2025.

[7] F. A. Jibon, A. Jamil Chowdhury, M. H. Miraz, H. H. Jin, M. U. Khandaker, S. Sultana, S. Nur, F. H. Siddiqui, A. Kamal, M. Salman, et al., “Sequential graph convolutional network and deeprnn based hybrid framework for epileptic seizure detection from EEG signal,” Digital Health, vol. 10, p. 20552076241249874,2024.

[8] R. Zhu et al., “Epileptic seizure prediction via multidimensional transformer and recurrent neural network fusion,” Journal of Translational Medicine, vol. 22, p. 895, 2024.

[9] S. R. Sheikh, Z. A. McKee, S. Ghosn, K.-S. Jeong, M. Kattan, R. C. Burgess, L. Jehi, and C. Y. Saab,“Machine learning algorithm for predicting seizure control after temporal lobe resection using peri-ictal electroencephalography,” Scientific Reports, vol. 14, no. 1, p. 21771, 2024.

[10] K. Jusseaume, Data preparation process to predicting age in seizure patients via deep learning with electroencephalogram recordings: a thesis in Computer Science. PhD thesis, University of Massachusetts Dartmouth, 2022.

[11] K. Khan, “Explainable IoT-based machine learning framework for EEG seizure detection using Random Forest, SVM, and KNN classifiers”, 2024.

[12] K. Jusseaume, “Deep learning recurrent neural networks for brain age prediction in seizure patients using the Temple University Abnormal EEG Corpus (TUAB),” Sensors, vol. 22, 2022.

[13] P. Akor, G. Enemali, U. Muhammad, R. R. Singh, and H. Larijani, "Hierarchical Deep Learning for Comprehensive Epileptic Seizure Analysis: From Detection to Fine-Grained Classification," *Information*, vol. 16, no. 7, p. 532, 2025.